IMPERIAL

IMPERIAL COLLEGE LONDON

DEPARTMENT OF MATHEMATICS

SECOND-YEAR GROUP RESEARCH PROJECT

# Analytical and Numerical Methods for Solving Acoustic Diffraction Problems

*Authors:*
Michael Williams (CID: 02385632)
Kane Rowley (CID: 02220050)
Dan Haywood (CID: 02404325)

*Supervisor:*
Ory Schnitzer

July 20, 2026

**Abstract**

This project aims to investigate the diffraction of plane waves from a finite plate, exploring both asymptotic (using the geometric theory of diffraction to piece together the solution of the Sommerfeld problem and the WKB method) and numerical methods (using indirect boundary element methods) to solve the problem, as well as looking at near-field and far-field behaviour of the solution wave. We also compare the solutions found in both methods, and explore extensions to our ideas to allow for solutions to more general problems.

# Contents


**1 Introduction** **3**

**2 Formulation of the problem** **6**

**3 Analytical and Asymptotic Methods** **8**

3.1 Motivation . . . 8

3.2 The Sommerfeld problem . . . 8

3.2.1 Formulation . . . 8

3.2.2 Working in parabolic-cylindrical coordinates . . . 9

3.2.3 The Dirichlet case . . . 11

3.2.4 The Neumann case . . . 11

3.3 Finite plate problem . . . 12

3.3.1 Geometric Theory of Diffraction . . . 12

3.3.2 Asympotic Matching . . . 12

3.3.3 Far-field conditions . . . 13

3.3.4 Amplitude Function . . . 14

3.3.5 Phase Function . . . 15

**4 Numerical Methods** **17**

4.1 Motivation . . . 17

4.2 Fundamental solutions . . . 17

4.2.1 Properties of Green functions . . . 17

4.2.2 Layer potentials . . . 18

4.3 Constructing the boundary integral equation . . . 20

4.3.1 The Dirichlet problem . . . 20

4.3.2 The Neumann problem . . . 21

4.4 Boundary element methods . . . 21

4.5 Verifying our solution . . . 24

4.5.1 Near-field behaviour . . . 24

4.5.2 Far-field behaviour . . . 24

4.5.3 Mesh Convergence Analysis . . . 24

4.6 Refinements and extensions to the formulation . . . 30

4.6.1 Solving for an arbitrary plate . . . 30

4.6.2 Solving for an arbitrary plane wave . . . 30

4.6.3 Refining the discretisation . . . 31

**5 Comparison and conclusion** **34**

5.1 Comparison . . . . . 34
5.2 Conclusion . . . . . 34

**A Appendix** **36**

# 1 Introduction

One of the more thoroughly investigated problems in physics is the problem of *wave diffraction* from an obstacle – be it a slit or a grating. Starting in 1660 with Francesco Maria Grimaldi's first observations on what would later come to be known as diffraction in his 1665 book *Physico-Mathesis de Lumine, Coloribus et Iride* [1], scientists have deeply explored the effects of interference of a wave by an obstacle as well as the interaction of multiple diffracted wavefronts.

In 1860, Hermann von Helmholtz devised a partial differential equation describing the motion of a time-independent wave – one which was adapted from d'Alembert's earlier wave equation [2], which moved the context of waves from physics into mathematics:

$$\nabla^2 u(\mathbf{r}) + k^2 u(\mathbf{r}) = 0, \tag{1.1}$$

where $\nabla^2$ indicates the Laplace operator and $k = \frac{2\pi}{\lambda}$ is the *wave number* of the wave, where $\lambda$ is the wavelength of $u$, our time-independent wave.

This equation came to be known as the *Helmholtz equation* – one which we aim to study in detail. The Helmholtz equation can be used to model a wide variety of phyiscal problems. For example – it is used quite frequently in modelling time-harmonic diffusion, which is useful in medical applications, as well as in the time-independent Schrödinger equation (see equation 2.5 in David J. Griffiths' *Introduction to Quantum Mechanics, 3rd edn.* [3]), which is paramount in the field of quantum mechanics.

Importantly for our project, we can use the Helmholtz equation to model acoustic waves, as well as electromagnetic waves, which are comprised of a sum of two solutions to the Helmholtz equation – given certain conditions such as that the field to be solved for is *monochromatic*, ie. all waves share a frequency.

Focusing specifically on the context of the Helmholtz equation in acoustics, the aim of this project is to investigate the behaviour of waves incident on a finite plate located at $y = 0, -1 \leq x \leq 1$ in two dimensions. For the purposes of our study, we take the incident wave to be a *plane wave*, which is a type of solution to the two-dimensional Helmholtz equation of the form

$$u^{(i)} = e^{ik(x\cos\theta + y\sin\theta)}, \tag{1.2}$$

where $0 \leq \theta < 2\pi$ is the *angle of propagation* of the wave, measured from the positive $x$-axis (see Figure 1).

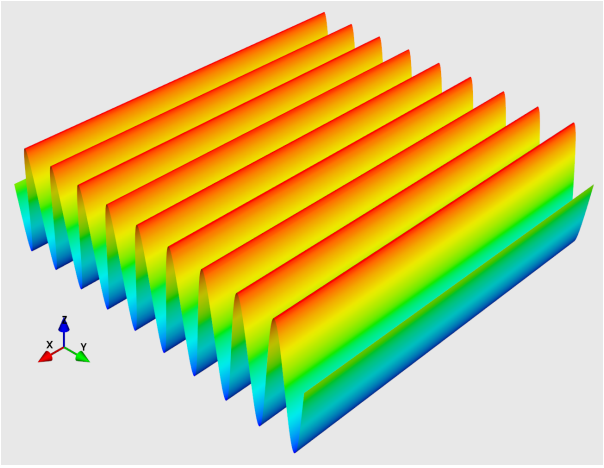


**Figure 1** A plot of the real part of the plane wave $e^{iky}$, with $k = 10$.

Solving problems of this type involves letting the total wave $u$ be represented as the sum of our known incident wave $u^{(i)}$ and an *unknown scattered wave* $u^{(s)}$, and determining $u^{(s)}$ from solving the differential equation governing the motion of the total wave. This can be done through a multitude of different methods – and this project aims to explore two main avenues of solving this problem:

**An analytical method** We first look at a more simplified case of the problem. Instead of our obstacle being a finite plate, we investigate the case where our obstacle is a *semi-infinite plate*, that is, a plate located at $y = 0$, $x \leq 0$. This problem is more commonly known as the *Sommerfeld problem*. Arnold Sommerfeld [4] showed in 1896 that it was possible to obtain an exact solution to this problem, as long as two conditions were met:

- The aptly named *Sommerfeld radiation condition*, which ensures that the scattered wave $u^{(s)}$ is *outward-propagting* – ie. that the obstacle is a *source* of waves as opposed to a *sink*.
- The *finite energy condition*, which ensures that when approaching edge of the plate, $x = y = 0$, the scattered wave $u^{(s)}$ does not grow too quickly, as this would indicate that energy is 'created' there – which does not make sense physically.

These conditions ensure that there is a *unique solution* to the Sommerfeld problem, and actually apply to any diffraction problem, so we will also enforce them in our finite plate problem.

To solve this problem, we employ a technique devised by Horace Lamb in the 1900s [5], which is to transform into parabolic-cylindrical coordinates, denoted by $(\xi, \eta, z)$, which causes the half-plane $(x \leq 0, y = 0)$ to now be represented by the line $\xi = 0$. This allows us to simplify the problem greatly by analysing the parabolic-cylindrical form of Helmholtz's equation.

Solving this problem gives us a solution in terms of Fresnel integrals, which are functions commonly used in the field of optics – used to model the field intensity of light interference from a corner, which is very similar to our problem. Using this representation, we move on to match this simpler solution to our finite plate problem. We do this using the *Geometric Theory of Diffraction* (GTD), introduced by Joseph Keller in 1953 [6], which models the behaviour of waves as rays given a high limit frequency. It explains how edge diffracted waves operate and why we get a smooth and continous transition from the incident region to the region behind the boundary, known as the shadow region. Using a method similar to the *Wentzel-Kramers-Brillouin (WKB) method* [6] we associate the ray with a field, $u(x, y) \sim A(x, y)e^{ikS(x,y)}$ and determine the coefficients $A$ and $S$ through matched asymptotic expansion. We formulate solutions using the conservation of energy and find the asymptotic behaviour of the Sommerfeld problem to determine the rays generated at the plate locally in the finite problem. This technique is known as the matched asymptotic expansion.

In Section 2 we precisely define the greater finite plate problem, before moving onto Section 3 where we go on to define the Sommerfeld problem, solve it exactly, and apply our results to the finite plate problem along with the GTD.

**A numerical method** We first look at the concept of *fundamental solutions* to our problem. In 1828, George Green [7] published his now famous *An Essay on the Application of Mathematical Analysis to the Theories of Electricity and Magnetism*, which describes the concept of what are now known as *Green functions*, which are solutions to the governing equation (in our case, the Helmholtz equation) at every point except a *source point* $\mathbf{r_0}$. These play an important role in our problem; as we show in Section 4, the Green functions of our

system (and hence any superposition of them) satisfy both the Helmholtz equation – by construction – as well as the Sommerfeld radiation condition.

We use this result to construct $u^{(s)}$ – we represent it as the superposition of a continuous distribution of these fundamental solutions along our obstacle, and use the notion of *jump relations*, which are discontinuities in this superposition when moving across the boundary, in order to form a *boundary integral equation* (BIE) to solve for our distribution, $\phi$.

We then solve this integral equation numerically, using what is known as a *boundary element method* (BEM), which is a relatively recent development in numerical computation – the first textbooks detailing such methods being published in the 1970s. This involves partitioning our boundary into discrete *boundary elements*, and letting the distribution $\phi$ be constant on each boundary element.

In Section 4 we also look at extensions to the numerical method used – such as how we can generalise the method to an arbitrary boundary, as well as refinements – such as adapting the BEM to work more favourably for our specific problem.

After detailing these methods, in Section 5 we compare the solutions obtained from each method, and discuss how the scheme can be further generlised as well as alternative methods to solving these types of problems. We also provide figures to give a more visual representation of our solutions, as well as the repository containing the programs used for the numerical approach (see Appendix A) – all plots and code used in this project are our own work.

The methods detailed in this paper are only a few of the many methods that can be used to solve these types of problems – for more examples of the various approaches one can take, the reader may be interested in looking at papers by Chen and Zhou [8], Colton and Kress [9], Noble [10], Crighton [11] and Born and Wolf [12].

## 2 Formulation of the problem

Consider the 2D case of the Helmholtz equation:

$$\nabla^2 u(x,y) + k^2 u(x,y) = 0, \tag{2.1}$$

where $u(x,y)$ indicates the total wave of the system. Suppose there is an infinitely thin, rigid plate located at the boundary defined by

$$\gamma = \{(x,y) : -1 \leq x \leq 1, y = 0\}.$$

Then, our system is subject to either of the two boundary conditions on $\gamma$:

$$\begin{cases} u(x,y) = 0 & \text{(Dirichlet conditions)} \\ \frac{\partial u(x,y)}{\partial n} = 0 & \text{(Neumann conditions)} \end{cases} \tag{2.2}$$

with $n$ indicating the direction of the normal of $\gamma$, which in our case is $y$.

In our context of acoustic waves, Dirichlet conditions on $\gamma$ imply that the plate is *sound-soft*, while Neumann conditions indicate that the plate is *sound-hard*. These two cases both imply that the obstacle is impenetrable by acoustic waves, however the diffraction and reflection produced will differ depending on the circumstance.

We split our total wave $u(x,y)$ into two parts: a *known incident wave* $u^{(i)}(x,y)$, and an *unknown scattered wave* $u^{(s)}(x,y)$, which is the wave generated at the obstacle $\gamma$. In our problem, we are looking at the case where $u^{(i)}(x,y) = e^{iky}$, which represents a *plane wave* travelling at an angle $\frac{\pi}{2}$ from the positive $x$-axis, normally incident on our plate. We can easily check that $u^{(i)}$ solves (2.1) – indeed $\nabla^2 u^{(i)} = -k^2 u^{(i)}$. Thus, using the superposition principle, $u$ solves our system if and only if $u^{(s)}$ solves the system with boundary conditions

$$\begin{cases} u^{(s)} = -u^{(i)} & \text{(Dirichlet conditions)} \\ \frac{\partial u^{(s)}}{\partial n} = -\frac{\partial u^{(i)}}{\partial n} & \text{(Neumann conditions).} \end{cases} \tag{2.3}$$

By itself, this system is not enough to guarantee that the solution $u(x,y)$ is *unique*. To secure the uniqueness of our solutions, we also enforce two conditions on our *scattered wave* $u^{(s)}$.

The first is the *Sommerfeld radiation condition*:

$$\lim_{r\to\infty} \sqrt{r}\left(\frac{\partial u^{(s)}}{\partial r} - iku^{(s)}\right) = 0, \tag{2.4}$$

where $r$ is the distance to the origin, $||(x,y)||$.

For the purposes of our problem, we will state this condition in a different form:

$$u^{(s)} \sim A(\theta)\frac{e^{ikr}}{\sqrt{r}}, \quad r \to \infty, \tag{2.5}$$

where $A(\theta)$ is a function independent of $r$, which will be analysed. We can easily see that $u^{(s)}$, in this form, satisfies (2.4).

The second is the *finite energy condition* at the edges, which states that $\exists\, C, \epsilon > 0$ :

$$\forall (\mathbf{r}_n)_{n\geq 0} \to \mathbf{r} \text{ with } ||\mathbf{r}_n - \mathbf{r}|| < \epsilon, |u^{(s)}(\mathbf{r}_n)| \leq C||\mathbf{r}_n - \mathbf{r}||^\alpha \tag{2.6}$$

with $\alpha > \frac{-1}{2}$.

This is derived from the requirement that in any small domain the edges of the boundary, the energy density of the scattered wave must be finitely integrable. [13]

Figure 2 shows an illustration of the problem setup, with 4 different sections outlined, where we expect to see different behaviour of the wave:

1. The area defined by $\{(x, y) : -1 < x < 1, y < 0\}$, where we expect to see reflection of $u^{(i)}$ as well as diffraction from the edges,
2. The area defined by $\{(x, y) : x < -1, y \in \mathbb{R}\}$, where we expect to see $u^{(i)}$ as well as some diffraction,
3. The area defined by $\{(x, y) : -1 < x < 1, y > 0\}$, where we expect to see a shadow region as well as a diffraction pattern,
4. The area defined by $\{(x, y) : x > 1, y \in \mathbb{R}\}$, which is similar to section 2.

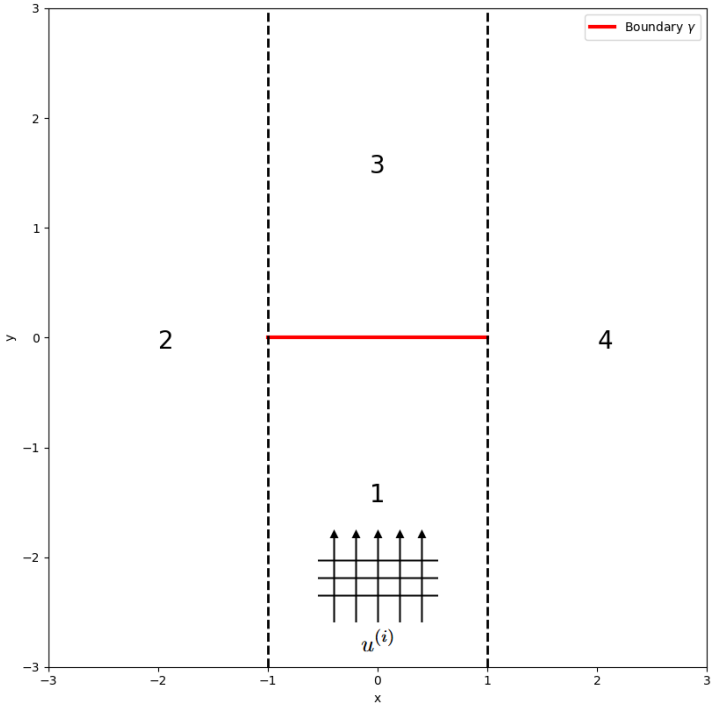



**Figure 2** A diagram of the setup of the finite plate problem, with the 4 sections outlined in Section 2 shown.

These conditions, together, allows us to find a unique solution $u^{(s)}$ to the system, and hence a unique total wave $u$.

# 3 Analytical and Asymptotic Methods

## 3.1 Motivation

Before we attempt to solve the finite plate problem, we first analytically solve the canonical Sommerfeld problem. We follow McDonald's method of using parabolic cylindrical coordinates. [14]

Afterwards, in the high frequency limit we asymptotically match this solution to the plate scale and use the geometric theory of diffraction to determine what waves are generated at the edges of the plate.

## 3.2 The Sommerfeld problem

### 3.2.1 Formulation

The Sommerfeld problem requires us to determine the total field of a wave, given that plane waves are normally incident upon a rigid, infinitely thin, semi-infinite screen (i.e. a half plane). [15]

Suppose we work in a new coordinate system that is of a certain length scale $(\tilde{x}, \tilde{y})$. Our boundary is defined as $\tilde{x} < 0, \tilde{y} = 0$ and has either a Dirichlet or Neumann boundary condition.

Incident waves of a certain frequency are of the form $\tilde{u}^{(i)}(\tilde{x}, \tilde{y}) = e^{i\tilde{y}}$. They travel from $\tilde{y} < 0$ and satisfy the 2D Helmholtz equation,

$$\nabla^2 \tilde{u}^{(i)}(\tilde{x}, \tilde{y}) + \tilde{u}^{(i)}(\tilde{x}, \tilde{y}) = 0. \tag{3.1}$$

The total wave, $\tilde{u}$ is decomposed into three parts, the incident, reflected and diffracted part $\tilde{u}_i$, $\tilde{u}_r$, $\tilde{u}_d$ respectively. We have three regions that have different decompositions – as shown in Figure 3:

$$\tilde{u} = \begin{cases} \tilde{u}^{(i)}(\tilde{x}, \tilde{y}) + \tilde{u}^{(r)}(\tilde{x}, \tilde{y}) + \tilde{u}^{(d)}(\tilde{x}, \tilde{y}) & \tilde{x} < 0, \tilde{y} < 0 \text{ (Incident + Reflected)} \\ \tilde{u}^{(d)}(\tilde{x}, \tilde{y}) & \tilde{x} < 0, \tilde{y} > 0 \text{ (Shadow Region)} \\ \tilde{u}^{(i)}(\tilde{x}, \tilde{y}) + \tilde{u}^{(d)}(\tilde{x}, \tilde{y}) & \tilde{x} > 0 \text{ (Incident Only)} \end{cases} \tag{3.2}$$

Consider the waves that are normally incident on the boundary (region 1). Express the total wave function as:

$$\tilde{u}(\tilde{x}, \tilde{y}) = \tilde{u}^{(i)}(\tilde{x}, \tilde{y}) + \tilde{u}^{(s)}(\tilde{x}, \tilde{y}), \tag{3.3}$$

where $\tilde{u}^{(s)}$ is the scattered wave from the boundary. Mathematician Lamb [16] surmised that the scattered wave of a Sommerfeld problem has the form:

$$\tilde{u}^{(s)}(\tilde{x}, \tilde{y}) = a(\tilde{x}, \tilde{y})e^{i\tilde{y}} + b(\tilde{x}, \tilde{y})e^{-i\tilde{y}}, \tag{3.4}$$

where $a, b$ are the amplitudes of the transmitted and reflected part of the wave respectively.

$\tilde{u}^{(s)}$ obeys Helmholtz's equation,

$$\nabla^2 \tilde{u}^{(s)}(\tilde{x}, \tilde{y}) + \tilde{u}^{(s)}(\tilde{x}, \tilde{y}) = 0. \tag{3.5}$$

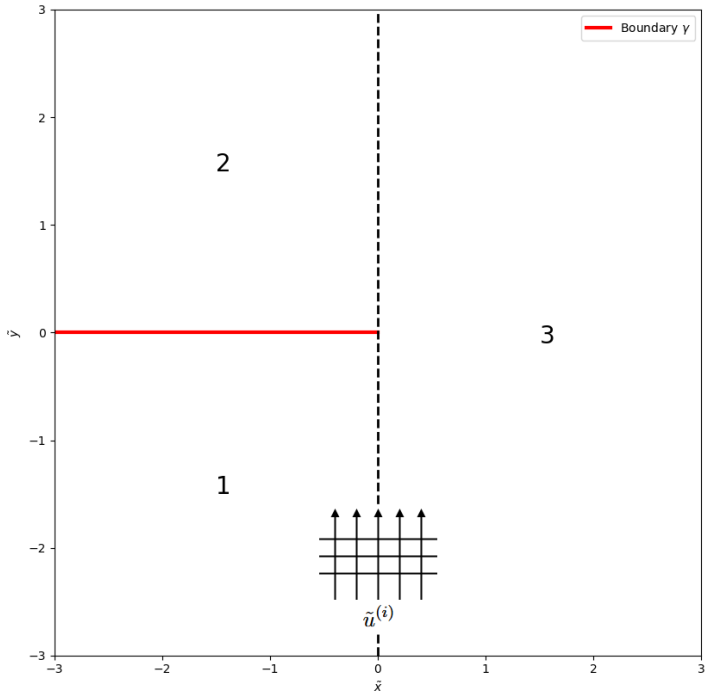



**Figure 3** A diagram of the setup of the Sommerfeld problem, with the 3 sections outlined in Section 3.2 shown.

### 3.2.2 Working in parabolic-cylindrical coordinates

To solve $\tilde{u}^{(s)}$, consider the parabolic-cylindrical coordinate $(\xi, \eta, z)$, related to both the cylindrical and Cartesian coordinates, $(\tilde{r}, \tilde{\phi}, z), (\tilde{x}, \tilde{y}, z)$ (for $0 \leq \tilde{\phi} \leq 2\pi$):

$$\begin{aligned} \tilde{x} + i\tilde{y} = \tilde{r}e^{i\tilde{\phi}} &= (\xi + i\eta)^2, \\ \tilde{x} = \xi^2 - \eta^2, \quad \tilde{y} &= 2\xi\eta \\ \xi = \sqrt{\tilde{r}}\cos\frac{\tilde{\phi}}{2}, \quad \eta &= \sqrt{\tilde{r}}\sin\frac{\tilde{\phi}}{2}. \end{aligned} \tag{3.6}$$

Where $\tilde{r} = \sqrt{\tilde{x}^2 + \tilde{y}^2}$ and $\tilde{\phi}$ is the polar angle in the $(\tilde{x}, \tilde{y})$ coordinate system. For convention, the sign of $\xi$ at a point $(\xi, \eta) = (\tilde{x}, \tilde{y})$ is the same as the sign of $\tilde{y}$, while $\eta$ is always non-negative. The half plane $(\tilde{x} > 0, \tilde{y} = 0)$ corresponds to $\eta = 0$, and the complementary half plane $(\tilde{x} < 0, \tilde{y} = 0)$ has coordinate $\xi = 0$. Using the chain rule we obtain,

$$\begin{aligned} \frac{\partial\tilde{u}}{\partial\xi} &= 2\xi\frac{\partial\tilde{u}}{\partial\tilde{x}} + 2\eta\frac{\partial\tilde{u}}{\partial\tilde{y}} \\ \frac{\partial^2\tilde{u}}{\partial\xi^2} &= 2\frac{\partial\tilde{u}}{\partial\tilde{x}} + 4\xi^2\frac{\partial^2\tilde{u}}{\partial\tilde{x}^2} + 8\eta\xi\frac{\partial^2\tilde{u}}{\partial\tilde{y}\tilde{x}} + 4\eta^2\frac{\partial^2\tilde{u}}{\partial\tilde{y}^2} \\ \frac{\partial\tilde{u}}{\partial\eta} &= -2\eta\frac{\partial\tilde{u}}{\partial\tilde{x}} + 2\xi\frac{\partial\tilde{u}}{\partial\tilde{y}} \\ \frac{\partial^2\tilde{u}}{\partial\eta^2} &= -2\frac{\partial\tilde{u}}{\partial\tilde{x}} + 4\eta^2\frac{\partial^2\tilde{u}}{\partial\tilde{x}^2} - 8\eta\xi\frac{\partial^2\tilde{u}}{\partial\tilde{y}\tilde{x}} + 4\xi^2\frac{\partial^2\tilde{u}}{\partial\tilde{y}^2}, \end{aligned} \tag{3.7}$$

Combining the equations, we can now express Helmholtz's equation in terms of parabolic-cylindrical coordinates:

$$\frac{\partial^2\tilde{u}^{(s)}}{\partial\xi^2} + \frac{\partial^2\tilde{u}^{(s)}}{\partial\eta^2} + 4(\xi^2 + \eta^2)\tilde{u}^{(s)} = 0 \tag{3.8}$$

We also take partial derivatives of $\tilde{u}^{(s)}(\xi,\eta)$ with respect to $\xi$ using (3.4):

$$\frac{\partial^2\tilde{u}^{(s)}}{\partial\xi^2}=\left(\frac{\partial^2 a}{\partial\xi^2}+4i\eta a\frac{\partial a}{\partial\xi}-4\eta^2 a\right)e^{2i\eta\xi}+\left(\frac{\partial^2 b}{\partial\xi^2}-4i\eta b\frac{\partial b}{\partial\xi}-4\eta^2 b\right)e^{-2i\eta\xi}. \tag{3.9}$$

The identity is symmetrical for $\frac{\partial^2\tilde{u}^{(s)}}{\partial\eta^2}$.

It follows that $a$ and $b$ obey the differential equations,

$$\frac{\partial^2(a,b)}{\partial\xi^2}+\frac{\partial^2(a,b)}{\partial\eta^2}\pm 4i\left(\eta\frac{\partial(a,b)}{\partial\xi}+\xi\frac{\partial(a,b)}{\partial\eta}\right)=0, \tag{3.10}$$

where the last term is postitive for $a$ and negative for $b$. Taking $a,b(\xi,\eta)=f(\xi\pm\eta)\equiv f(\zeta_{a,b})$, (3.10) becomes

$$f''+2i\zeta_{a,b}f'=0. \tag{3.11}$$

Solve the ODE problem for $a$ and $b$,

$$f(\zeta_{a,b})=A+B\int_0^{\zeta_{a,b}}e^{-i\zeta^2}\,d\zeta, \tag{3.12}$$

where $A$ and $B$ are constants. Hence,

$$\begin{aligned}a=f(\zeta_u)&=C_a+D_a\int_0^{\xi+\eta}e^{-i\zeta^2}\,d\zeta,\\ b=f(\zeta_v)&=C_b+D_b\int_0^{\xi-\eta}e^{-i\zeta^2}\,d\zeta,\end{aligned} \tag{3.13}$$

and,

$$\tilde{u}^{(s)}=C_a e^{i\tilde{y}}+C_b e^{-i\tilde{y}}+D_a e^{i\tilde{y}}\int_0^{\xi+\eta}e^{-i\zeta^2}d\zeta+D_b e^{-i\tilde{y}}\int_0^{\xi-\eta}e^{-i\zeta^2}d\zeta. \tag{3.14}$$

To obtain $C_a, D_a, C_b, D_b$, note that if $\tilde{x}\to\infty$ and $\tilde{y}$ is small ($\xi\to-\infty$ and $\eta$ is small), the scattered wave is negligible. Hence we must separately have,

$$C_a+D_a\int_0^{\xi}e^{-i\zeta^2}d\zeta\to 0,\quad C_b+D_b\int_0^{\xi}e^{-i\zeta^2}d\zeta\to 0. \tag{3.15}$$

Using the fact that [17]:

$$\int_0^{\infty}e^{-i\zeta^2}d\zeta=\frac{\sqrt{\pi}}{2}e^{-i\pi/4}, \tag{3.16}$$

We obtain,

$$C_a+\tfrac{\sqrt{\pi}}{2}D_a e^{-i\pi/4}=0,\quad C_b+\tfrac{\sqrt{\pi}}{2}D_b e^{-i\pi/4}=0. \tag{3.17}$$

### 3.2.3 The Dirichlet case

Our wave vanishes (amplitude is zero) along the semi-infinite boundary, $\tilde{x} < 0, \tilde{y} = 0$ which corresponds to $\xi = 0$. This implies:

$$-1 = \tilde{u}^s(\tilde{x} < 0, 0) = \tilde{u}^s(0, \eta) = C_a + C_b + (D_a - D_b) \int_0^\eta e^{-i\zeta^2}\, d\zeta \tag{3.18}$$

This is valid for any $\tilde{x} < 0$. Take $\tilde{x} \to 0^-$. Then

$$C_a + C_b = -1, \quad D_b = D_a. \tag{3.19}$$

Combine this with (3.17) to obtain

$$C_a = C_b = -1/2, \;\; D_a = D_b = \pi^{-1/2} e^{i\pi/4} \tag{3.20}$$

We obtain the solution in terms of Fresnel Integrals,

$$\tilde{u}^{(s)} = \frac{e^{i(\tilde{y}+\pi/4)}}{\sqrt{\pi}} \int_0^{\xi+\eta} e^{-i\zeta^2} d\zeta + \frac{e^{-i(\tilde{y}-\pi/4)}}{\sqrt{\pi}} \int_0^{\xi-\eta} e^{-i\zeta^2} d\zeta - \frac{e^{i\tilde{y}} + e^{-i\tilde{y}}}{2} \tag{3.21}$$

### 3.2.4 The Neumann case

Our boundary condition has changed to:

$$\frac{\partial \tilde{u}^{(s)}(\tilde{x} < 0, 0)}{\partial \tilde{y}} = \frac{\partial \tilde{u}^{(s)}(0, \eta)}{\partial \tilde{y}} = -i. \tag{3.22}$$

We use relation,

$$\frac{\partial \xi}{\partial \tilde{y}} = \frac{\eta}{2\tilde{r}}, \quad \frac{\partial \eta}{\partial \tilde{y}} = \frac{\xi}{2\tilde{r}}. \tag{3.23}$$

We find,

$$\frac{\partial \tilde{u}^{(s)}(\tilde{x} < 0, 0)}{\partial \tilde{y}} = \frac{\partial \tilde{u}^{(s)}(0, \eta)}{\partial \tilde{y}} = i(C_a - C_b) + (D_a + D_b)\left( i \int_0^\eta e^{-i\zeta^2} d\zeta + \frac{1}{2\eta} e^{-i\eta^2} \right) = -i. \tag{3.24}$$

Using (3.17) we obtain

$$C_a = -C_b = -\frac{1}{2}, D_a = -D_b = \frac{1}{\sqrt{\pi}} e^{i\pi/4}. \tag{3.25}$$

The solution is thus

$$\tilde{u}^{(s)} = \frac{e^{i(\tilde{y}+\pi/4)}}{\sqrt{\pi}} \int_0^{\xi+\eta} e^{-i\zeta^2} d\zeta - \frac{e^{-i(\tilde{y}-\pi/4)}}{\sqrt{\pi}} \int_0^{\xi-\eta} e^{-i\zeta^2} d\zeta - \frac{e^{i\tilde{y}} - e^{-i\tilde{y}}}{2} \tag{3.26}$$

## 3.3 Finite plate problem

### 3.3.1 Geometric Theory of Diffraction

We return to our finite plate problem formulated in Figure 2. In the finite problem, it is possible for diffracted waves to interact with the opposite edge to produce a new source of diffraction. This can cause a series of diffraction to occur making the analytical approach to the problem more difficult and unviable. To avoid this issue, we seek for an asymptotic solution for the scattered wave in the case where waves have a high frequency limit. This motivates the following theorem.

**Theorem 3.1** (Geometric Theory of Diffraction)**.** *[18]*

*Suppose a wave travels through a homogeneous medium in which the wavelength is small compared to the length of the boundary ($2 \gg \lambda$ or $k \gg \frac{1}{2}$).*

*The incident, reflected and diffracted waves propagate along rays. All three waves travel in straight lines and for the edge-diffracted rays, the angle between the diffracted ray and the edge is equal to the angle between the incident ray and the edge, when both rays lie in the same medium. Furthermore, the diffracted ray lies on the opposite side of the normal plane from the incident ray.*

It follows that an incident ray produces a cone of edge diffracted rays when it hits the edge, see Figure 4. We use this fact to approximate the local scattered solution around the edge in a high frequency limit.

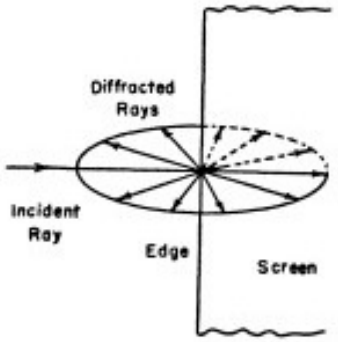



**Figure 4** An incident ray hitting an edge creates a cone of diffracted rays, as described by the Geometric Theory of Diffraction (GTD). [18]

### 3.3.2 Asympotic Matching

In order to understand how the diffracted waves behave locally, we make use of the solution to the Sommerfeld problem. Using 3.2, we attempt to 'match' the coordinate system from the Sommerfeld problem to the plate scale in the finite plate problem. By observing the asymptotic behaviour of the Sommerfeld problem, this technique enables us to determine what rays are generated at the plate locally.

We are interested in the region that is a distance of wavelength order away from the edge of the plate, i.e. $O(\frac{1}{k})$. Hence, we use the following transformation of coordinates,

$$(\tilde{x}, \tilde{y}) = (k(x-1), ky). \tag{3.27}$$

Assuming the high frequency limit $k \to \infty$, the coordinate system has been stretched such that the edge of the plate $(1, 0)$ is transformed into the edge of the Sommerfeld problem, whilst keeping $k$ fixed.

It follows incident rays of order $\frac{1}{k}$ close to the edge can be expressed as $u^{(i)}(x, y) = \tilde{u}^{(i)}(\tilde{x}, \tilde{y}) = e^{i\tilde{y}}$ and satisfies 2D Helmholtz equation (3.1). We repeat the same steps as 3.2 to obtain an analytical solution for both Dirichlet and Neumann boundary conditions.

### 3.3.3 Far-field conditions

We have found an analytical wave solution to the Sommerfeld problem in terms of Fresnel integrals. We now wish to observe the solution asymptotically to determine the far-field conditions of the diffracted wave $\tilde{u}^{(d)}$ as $\tilde{r} \to \infty$ keeping $k$ fixed.

We expect the far-field condition to follow the 2D Sommerfeld radiation condition,

$$\lim_{\tilde{r}\to\infty} \sqrt{\tilde{r}} \left( \frac{\partial \tilde{u}}{\partial \tilde{r}} - i\tilde{u} \right) = 0. \tag{3.28}$$

resulting in outward wave propagation.

We will inspect the far-field conditions separately for the 3 regions (3.2).

**Region 1 ($\tilde{x} < 0, \tilde{y} < 0$)**

Incident rays in this region will hit normal to the plate, so we expect to see reflected waves of the form, $u^{(r)} = Ae^{-iky}$. For the Dirichlet boundary case, $A = -1$ and for the Neumann boundary case, $A = 1$.

Recall that,

$$\tilde{u}(\tilde{x}, \tilde{y}) = \tilde{u}^{(i)}(\tilde{x}, \tilde{y}) + \tilde{u}^{(r)}(\tilde{x}, \tilde{y}) + \tilde{u}^{(d)}(\tilde{x}, \tilde{y}) \tag{3.29}$$

where $\tilde{u}^{(d)}$ is the edge diffracted part of the wave (edge located $(\tilde{x} = 0, \tilde{y} = 0)$).

We work with the Dirichlet case first. Removing the reflected ray from (3.21) we obtain,

$$\tilde{u}^{(d)} = \frac{e^{i(\tilde{y}+\pi/4)}}{\sqrt{\pi}} \int_0^{\xi+\eta} e^{-i\zeta^2} d\zeta + \frac{e^{-i(\tilde{y}-\pi/4)}}{\sqrt{\pi}} \int_0^{\xi-\eta} e^{-i\zeta^2} d\zeta - \frac{e^{i\tilde{y}} - e^{-i\tilde{y}}}{2} \tag{3.30}$$

Using integration by parts, the first order asymptotic approximation for the Fresnel integral $F(X) = \int_0^X e^{-i\zeta^2} d\zeta$ when $|X|$ is large [15] is,

$$F(X) \approx \operatorname{sgn}(X) \cdot \frac{\sqrt{\pi}}{2} e^{-i\pi/4} - \frac{i}{2X} e^{-iX^2}$$

In $\tilde{x} < 0, \tilde{y} < 0$, $\pi < \tilde{\phi} < \frac{3\pi}{2}$. From (3.6), we get $\operatorname{sgn}(\xi + \eta) = 1$ and $\operatorname{sgn}(\xi - \eta) = -1$. Hence,

$$\tilde{u}^{(d)} \approx \frac{e^{i(\tilde{y}+\pi/4)}}{\sqrt{\pi}} \left[ \frac{-i}{2(\xi+\eta)} e^{-i(\xi+\eta)^2} \right] + \frac{e^{-i(\tilde{y}-\pi/4)}}{\sqrt{\pi}} \left[ \frac{-i}{2(\xi-\eta)} e^{-i(\xi-\eta)^2} \right] \tag{3.31}$$

We simplify the equation and convert to polar coordinates,

$$\tilde{u}^{(d)} \sim \frac{e^{-i\tilde{r}}}{\sqrt{\tilde{r}}} \left( \frac{-ie^{i\pi/4} \cos\frac{\tilde{\phi}}{2}}{\sqrt{\pi} \cos\tilde{\phi}} \right) = A(\tilde{\phi}) \frac{e^{-i\tilde{r}}}{\sqrt{\tilde{r}}}, \tilde{r} \to \infty \tag{3.32}$$

$\tilde{u}^{(d)}$ is proportional to $\frac{1}{\sqrt{\tilde{r}}}$ so we expect to see the amplitude decay as we travel away from the source. Clearly our solution satisfies the Sommerfeld radiation condition (3.28) and the wave is outward propagating as $\tilde{r} \to \infty$.

Similarly for the Neumann case we have,

$$\tilde{u}^{(d)} \sim \frac{e^{-i\tilde{r}}}{\sqrt{\tilde{r}}} \left( \frac{ie^{i\pi/4} \sin \frac{\tilde{\phi}}{2}}{\sqrt{\pi} \cos \tilde{\phi}} \right) = B(\tilde{\phi}) \frac{e^{-i\tilde{r}}}{\sqrt{\tilde{r}}}, \tilde{r} \to \infty \tag{3.33}$$

**Region 2 ($\tilde{x} < 0, \tilde{y} > 0$)**

Both incident and reflected waves are not present, i.e. $\tilde{u} = \tilde{u}^{(d)} = \tilde{u}^{(s)}$ in (3.21). $\frac{\pi}{2} < \tilde{\phi} < \pi$, so $\text{sgn}(\xi + \eta) = 1$ and $\text{sgn}(\xi - \eta) = 1$. The terms cancel and we achieve (3.32) and (3.33) hence wave is outward propagating as $\tilde{r} \to \infty$.

**Region 3 ($\tilde{x} > 0, \tilde{y} \in \mathbb{R}$)**

No reflected waves are present in this region, just the incident wave. So $\tilde{u}^{(d)} = \tilde{u}^{(s)}$ in (3.21). In the Sommerfeld problem, $0 < \tilde{\phi} < \frac{\pi}{2}$ or $\frac{3\pi}{2} < \tilde{\phi} < 2\pi$.

$\text{sgn}(\xi + \eta) = 1$ and $\text{sgn}(\xi - \eta) = 1$. Like region 2, the exponential terms cancel and achieve the same expression as (3.32) and (3.33) hence same result.

Note for all regions we obtain the same diffraction constant $A(\tilde{\phi})$ and $B(\tilde{\phi})$ which we will apply to our local solution to our finite plate problem.

### 3.3.4 Amplitude Function

We proceed to find the asymptotic solution of the diffracted wave following J.B. Keller's method [18]. From 3.3.1, diffracted rays all emanate from the edges. Define one of the edge as a source point, $p_d = (1, 0)$. Each diffracted ray that start at $p_d$ will pass through some point $p = (x, y)$. Pick one diffracted ray from this source, say ray j with wave function $u_j$

Next, we associate this ray with a field, $u_j(x, y) \sim A_j(x, y)e^{ikS_j(x,y)}$ where $A_j(x, y)$ is the amplitude function and $S_j(x, y)$ is the phase function to be found. We first determine the amplitude function.

Suppose points $p, p' \in \mathbb{R}^2$ are on the ray. Think of a narrow tube of rays from $p$ to $p'$ where $d\sigma(p)$ and $d\sigma(p')$ are the corresponding cross sectional area of the tube containing this ray. As the power radiated through both areas must be the same, by the conservation of energy,

$$nA_j^2 d\sigma|_p = nA_j^2 d\sigma|_{p'}, \tag{3.34}$$

where n(q) is the refractive index at q. We are working in a homogeneous medium so let n(q) = 1 at all points $q \in \mathbb{R}^2$. It follows that

$$A_j(p) = \left[ \frac{d\sigma(p')}{d\sigma(p)} \right]^{1/2} A_j(p'). \tag{3.35}$$

Notice this is the solution to the linear transport equation of the WKB method for the same field approximation.

Rays in our finite plate problem are in two dimensions and by the geometric theory of diffraction, they are straight lines. From $p_d$ to $p$, the cross sectional area of the sector shaped strip is $d\sigma(x) = rd\theta$. $\theta$ is the angle subtended by the two bounding rays that intersect $p_d$ and $r$ is the distance between $p_d$ and $p$. Similarly, $d\sigma(p') = r'd\theta$. We obtain,

$$A_j(p) = \left(\frac{r'}{r}\right)^{1/2} A_j(p'). \tag{3.36}$$

Notice when $r' \to 0$, since the amplitude $A_j(p')$ becomes infinite in such a way that $(r')^{1/2} A_j(p')$ is constant. We must require that the sum of the diffracted and incident fields at the edge to satisfy the boundary conditions asymptotically. Hence we assume that this constant is proportional to the incident amplitude, $A_{inc}(p_d)$ at the point of diffraction.

$$(r')^{1/2} A_{\text{j}}(p') = D A_{\text{inc}}(p_d), \tag{3.37}$$

where D is the diffraction constant that depends on the angle of diffraction and boundary conditions of the edge. Notice D has dimensions $(\text{length})^{1/2}$. Hence D can be expressed as $D = k^{-1/2}\tilde{D}$ where $\tilde{D}$ is the dimensionless diffraction coefficient. This coefficient is already found from the canonical Sommerfeld problem as $A(\tilde{\phi})$ and $B(\tilde{\phi})$ through matched asymptotic expansion in (3.3.3).

Therefore the amplitude function (for Dirichlet boundary condition) is,

$$A_{\text{j}}(x, y) = \left(\frac{-ie^{i\pi/4}\cos\frac{\tilde{\phi}}{2}}{\sqrt{\pi}\cos\tilde{\phi}}\right)\frac{A_{\text{inc}}(x_d)}{\sqrt{kr}}, \tag{3.38}$$

where $\tilde{\phi} = \arctan\left(\frac{y}{x-1}\right)$ and $r = \sqrt{(x-1)^2 + y^2}$.

### 3.3.5 Phase Function

The phase function $S_j$ is defined as the optical length of ray j from $p$ to the source of the incident ray, point $p_0$:

$$S_{\text{diff}}(p) = \int_{p_0}^{p} ds = \int_{p_0}^{p_d} ds + \int_{p_d}^{p} ds = S_{\text{inc}}(p_d) + r. \tag{3.39}$$

$S_{\text{inc}}(p_d)$ is the phase of the incident wave at the point of diffraction. As the incident wave is approaching $p_d$ from $y < 0$ and is normal to the plate, $S_{\text{inc}}(p_d) = 0$.

Simplify the expression to obtain the asymptotic solution of the diffracted field.

$$\begin{aligned} u_{\text{diff}}(x) &\sim A_{\text{diff}}(x) e^{ikS_{\text{diff}}(x)} \\ &= \frac{\tilde{D} A_{\text{inc}}(x_d)}{(kr)^{1/2}} e^{ikS_{\text{inc}}(x_d)+ikr} \sim \left(\frac{-ie^{i\pi/4}\cos\frac{\tilde{\phi}}{2}}{\sqrt{\pi}\cos\tilde{\phi}}\right)\frac{e^{ikr}}{\sqrt{kr}} \end{aligned} \tag{3.40}$$

Converting to cartesian coordinates, the Dirichlet solution is,

$$u^{(d)}(x, y) \sim \frac{-ie^{i\pi/4}}{\sqrt{2\pi k}(x-1)}\left(\sqrt{\sqrt{(x-1)^2+y^2}+x-1}\right)e^{ik\sqrt{(x-1)^2+y^2}} \tag{3.41}$$

Similarly, the Neumann solution is:

$$u^{(d)}(x, y) \sim \frac{ie^{i\pi/4}}{\sqrt{2\pi k}(x-1)}\left(\sqrt{\sqrt{(x-1)^2+y^2}-x+1}\right)e^{ik\sqrt{(x-1)^2+y^2}} \tag{3.42}$$

Using symmetry, replace $x - 1$ with $x + 1$ to find the asymptotic solution for diffracted rays from the second source $(-1, 0)$.

Our final solution is the sum of the fields on all the rays through $(x, y)$. As $r \to \infty$,

$$u(x,y) \sim u^{(d)}_{(1,0)}(x,y) + u^{(d)}_{(-1,0)}(x,y) + u^{(i)}(x,y) + u^{(r)}(x,y) \tag{3.43}$$

Looking at Figure 2, only region 1 contains both $u^{(i)}(x,y)$ and $u^{(r)}(x,y)$, regions 2 and 4 contain $u^{(i)}(x,y)$ and region 3 contains neither. All regions contain diffracted rays from both edges.

# 4 Numerical Methods

## 4.1 Motivation

Moving back to the problem outlined in Section 2, we instead look at ways to solve for $u^{(s)}$ using numerical methods. Specifically, by representing $u^{(s)}$ as a distribution of wave point sources along $\gamma$, we can form a *boundary integral equation*, which can then be solved by employing a *boundary element method*. We will also address refinements and extensions that can be made to our model – such as how we may use the same method to solve for an arbitrarily placed plate, an arbitrarily angled wave, and an arbitrary number of plates.

## 4.2 Fundamental solutions

### 4.2.1 Properties of Green functions

One tool that we will use in the following sections is the notion of a *fundamental solution* to our problem. To investigate solutions for arbitrary problems we first need to establish analytical solutions to the most basic form of the problem – the one with no obstacle at all.

Specifically, we are looking for a function that satisfies the partial differential equation

$$\nabla^2 G(\mathbf{r}, \mathbf{r_0}) + k^2 G(\mathbf{r}, \mathbf{r_0}) = \delta(\mathbf{r} - \mathbf{r_0}), \tag{4.1}$$

where $\mathbf{r}, \mathbf{r_0} \in \mathbb{R}^2$ indicate position vectors $\mathbf{r} = (x, y)$, $\mathbf{r_0} = (a, b)$, as clearly this function satisfies (2.1) at every point except $\mathbf{r_0}$.

The solution to this equation $G(\mathbf{r}, \mathbf{r_0})$ is known as a *fundamental solution* or *Green function* for the problem.

For our equation, we get that

$$G(\mathbf{r}, \mathbf{r_0}) = \frac{i}{4} H_0^{(1)}(k||\mathbf{r} - \mathbf{r_0}||), \tag{4.2}$$

where $H_0^{(1)}(z)$ is the $0$-order Hankel function of the first kind.

The proof of this is outside the scope of this project – we will focus more on the applications of the fundamental solution – but if the reader is interested they may read more about this, as well as the extension to the 3D case, in G. Chen and J. Zhou's *Boundary element methods with applications to nonlinear problems*, pages 308–312. [8]

The most important property of $G(\mathbf{r}, \mathbf{r_0})$ is that it satisfies the Sommerfeld radiation condition, which is what allows us to construct our scattered wave $u^{(s)}$.

We can easily see this by looking at the large-$x$ asymptotic form for $H_0^{(1)}(x)$:

$$H_0^{(1)}(x) \sim \sqrt{\frac{2}{\pi x}} e^{i(x - \frac{\pi}{4})}, \tag{4.3}$$

where $x \in \mathbb{R}$. [19]

Thus, we have that, as $||\mathbf{r}|| = r \to \infty$,

$$\begin{aligned} G(\mathbf{r}, \mathbf{r_0}) &= \frac{i}{4} H_0^{(1)}(k||\mathbf{r} - \mathbf{r_0}||) \\ &\sim \frac{i}{4} \sqrt{\frac{2}{\pi k r}} e^{ikr} \\ &= A(\theta) \frac{e^{ikr}}{r^{\frac{1}{2}}}, \end{aligned} \tag{4.4}$$

with $A(\theta) = \sqrt{\frac{1}{8k\pi}}i, \quad 0 \leq \theta < 2\pi$, assuming that $\mathbf{r_0}$ is fixed.

In our specific application, $G(\mathbf{r}, \mathbf{r_0})$ corresponds to a *point source* of waves at $\mathbf{r_0}$, as shown in Figure 5. By the superposition principle, linear combinations of point sources centred at points on our boundary are also solutions to (2.1) and satisfy (2.5), as for large $||\mathbf{r}||$, the difference in $\mathbf{r_0}$ becomes negligible. Thus, we construct $u^{(s)}$ by taking it to be a continuous distribution of fundamental solutions of our problem.

### 4.2.2 Layer potentials

In order to construct $u^{(s)}$, we need to first define *single-* and *double-layer potentials*.

The *single-layer potential* is a continuous distribution of point sources $G(\mathbf{r}, \mathbf{r_0})$ along our obstacle $\gamma$. The desnsity of this distribution, which we denote as $\phi(\mathbf{r})$, is important as our plan is to impose our boundary conditions onto the system, which creates a boundary integral equation (BIE) for $\phi$, which, after solving, lets us obtain $u^{(s)}$.

Thus, we define the *single-layer potential* as

$$v_s(\mathbf{r}) = \int_\gamma \phi(\mathbf{r_0}) G(\mathbf{r}, \mathbf{r_0}) d\gamma(\mathbf{r_0}), \quad \mathbf{r} \in \mathbb{R}^2, \tag{4.5}$$

where $\phi(\mathbf{r})$ is our density function.

In a similar manner, we also define the *double-layer potential* as

$$v_d(\mathbf{r}) = \int_\gamma \phi(\mathbf{r_0}) \frac{\partial G(\mathbf{r}, \mathbf{r_0})}{\partial n(\mathbf{r_0})} d\gamma(\mathbf{r_0}), \quad \mathbf{r} \in \mathbb{R}^2, \tag{4.6}$$

where

$$\frac{\partial G(\mathbf{r}, \mathbf{r_0})}{\partial n(\mathbf{r_0})} = \mathbf{n}(\mathbf{r_0}) \cdot \nabla_{\mathbf{r_0}} G(\mathbf{r}, \mathbf{r_0}) = \mathbf{n}(\mathbf{r_0}) \cdot \frac{-ik(\mathbf{r} - \mathbf{r_0})}{4||\mathbf{r} - \mathbf{r_0}||} H_1^1(k||\mathbf{r} - \mathbf{r_0}||) \tag{4.7}$$

describes a *dipole* source at $\mathbf{r_0}$, as shown in Figure 6, which is made up of two equal and opposite sources, both lying on the line in the direction $\mathbf{n}(\mathbf{r_0})$ (which in our case is just the unit vector $\mathbf{j}$).

The single- and double-layer potentials, due to the superposition principle, both solve (2.1) and (2.5), and are also infinitely differentiable on $\mathbb{R}^2 \setminus \gamma$. [20]

This is desirable, as all we must do now is impose our boundary conditions, and solve for $\phi(\mathbf{r})$.

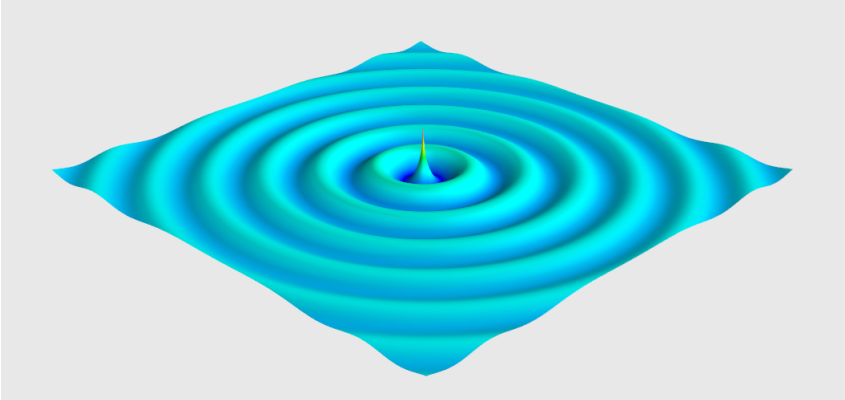


**Figure 5** A plot of the real part of $G(\mathbf{r}, 0)$. Notice how the waves' amplitude decays as $||\mathbf{r}||$ increases. This indicates that the wave satisfies the Sommerfeld radiation condition.

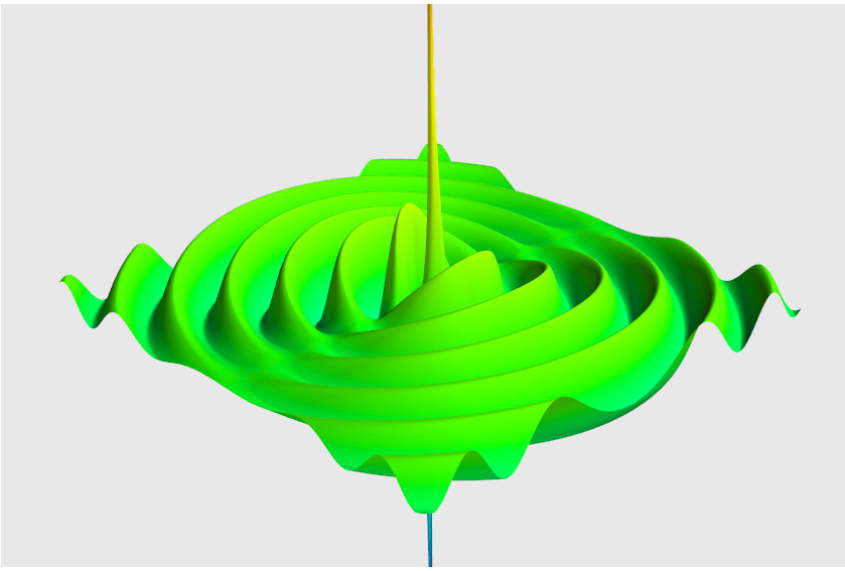


**Figure 6** A plot of the real part of $\frac{\partial G(\mathbf{r},\mathbf{r_0})}{\partial y(\mathbf{r_0})}$ at $\mathbf{r_0} = (0, 0)$. Again, the waves' amplitude decays as $||\mathbf{r}||$ increases, indicating that the wave satisfies the Sommerfeld radiation condition.

## 4.3 Constructing the boundary integral equation

### 4.3.1 The Dirichlet problem

In this section we focus specifically on the first of the two boundary conditions in (2.3):

$$u^{(s)} = -u^{(i)} = -1. \tag{4.8}$$

Our ansatz for $u^{(s)}$ will be a combination of the single- and double-layer potentials (4.5) and (4.6), as described in H. Brakhage and P. Werner's *Über das Dirichletsche Außenraumproblem für die Helmholtzsche Schwingungsgleichung*, pages 325–326: [21]

$$\begin{aligned} u^{(s)}(\mathbf{r}) &= v_d(\mathbf{r_0}) - iv_s(\mathbf{r_0}) \\ &= \int_\gamma \phi(\mathbf{r_0}) \frac{\partial G(\mathbf{r}, \mathbf{r_0})}{\partial n(\mathbf{r_0})} d\gamma(\mathbf{r_0}) - i \int_\gamma \phi(\mathbf{r_0}) G(\mathbf{r}, \mathbf{r_0}) d\gamma(\mathbf{r_0}) \\ &= \int_\gamma \phi(\mathbf{r_0}) \left( \frac{\partial G(\mathbf{r}, \mathbf{r_0})}{\partial n(\mathbf{r_0})} - iG(\mathbf{r}, \mathbf{r_0}) \right) d\gamma(\mathbf{r_0}). \end{aligned} \tag{4.9}$$

The most crucial theorem we will use to formulate our BIE is the following [9]:

**Theorem 4.1** (Jump relations)**.** *Suppose that the boundary $\partial\Omega$ of a set $\Omega \subset \mathbb{R}^2$ is sufficiently smooth and the function $\phi : \mathbb{R}^2 \to \mathbb{C}$ is continuous on $\partial\Omega$. Then, after defining $v_\pm(\mathbf{r}) = \lim_{h\to 0^+} v(\mathbf{r} \pm h\mathbf{n}(\mathbf{r}))$:*

1. *The value of the single-layer potential $v_s$ is continuous when moving across the boundary, ie. $v_{s+}(\mathbf{r}) = v_{s-}(\mathbf{r})$.*
2. *The normal derivative of the single-layer potential is well-defined on $\mathbb{R}^2 \setminus \partial\Omega$, but its value jumps when moving across the boundary, with:*

$$\begin{aligned} \frac{\partial v_{s\pm}(\mathbf{r})}{\partial n} &= \lim_{h\to 0^+} \mathbf{n}(\mathbf{r}) \cdot \nabla v_s(\mathbf{r} \pm h\mathbf{n}(\mathbf{r})) \\ &= \int_\gamma \phi(\mathbf{r_0}) \frac{\partial G(\mathbf{r}, \mathbf{r_0})}{\partial n(\mathbf{r})} d\gamma(\mathbf{r_0}) \pm \frac{1}{2}\phi(\mathbf{r}) \end{aligned} \tag{4.10}$$

3. *The double-layer potential $v_d$ is well-defined on $\mathbb{R}^2 \setminus \partial\Omega$, but its value jumps when moving across the boundary, with:*

$$\begin{aligned} v_{d\pm}(\mathbf{r}) &= \lim_{h\to 0^+} v_d(\mathbf{r} \pm h\mathbf{n}(\mathbf{r})) \\ &= \int_\gamma \phi(\mathbf{r_0}) \frac{\partial G(\mathbf{r}, \mathbf{r_0})}{\partial n(\mathbf{r_0})} d\gamma(\mathbf{r_0}) \mp \frac{1}{2}\phi(\mathbf{r}) \end{aligned} \tag{4.11}$$

4. *The value of the normal derivative of the double-layer potential is continuous when moving across the boundary, ie. $\frac{\partial v_{d+}(\mathbf{r})}{\partial n} = \frac{\partial v_{d-}(\mathbf{r})}{\partial n}$.*

Using these identities, we can construct a BIE for $\phi(\mathbf{r})$ with Dirichlet conditions:

On $\gamma$, we have:

$$\begin{aligned} u^{(s)} &= v_{d-}(\mathbf{r_0}) - iv_s(\mathbf{r_0}) \\ &= \int_\gamma \phi(\mathbf{r_0}) \frac{\partial G(\mathbf{r}, \mathbf{r_0})}{\partial n(\mathbf{r_0})} d\gamma(\mathbf{r_0}) + \frac{1}{2}\phi(\mathbf{r}) - i \int_\gamma \phi(\mathbf{r_0}) G(\mathbf{r}, \mathbf{r_0}) d\gamma(\mathbf{r_0}) \\ &= -1 \end{aligned} \tag{4.12}$$

where we used the jump relation (4.11). Rearranging terms then gives us the BIE

$$\frac{1}{2}\phi(\mathbf{r}) = -1 + \int_\gamma \phi(\mathbf{r_0}) \left( iG(\mathbf{r}, \mathbf{r_0}) - \frac{\partial G(\mathbf{r}, \mathbf{r_0})}{\partial n(\mathbf{r_0})} \right) d\gamma(\mathbf{r_0}), \tag{4.13}$$

where we now must solve for $\phi(\mathbf{r})$.

### 4.3.2 The Neumann problem

Now, we shift our focus to the second of the two boundary conditions in (2.3):

$$\frac{\partial u^{(s)}}{\partial n} = -\frac{\partial u^{(i)}}{\partial n} = -ik. \tag{4.14}$$

Again, our ansatz is a combination of single- and double-layer potentials as in (4.9), and our goal is again to formulate a BIE by imposing our boundary conditions on the system and using the jump relation to form an integral equation.

This time on $\gamma$ we have:

$$\begin{aligned} \frac{\partial u^{(s)}}{\partial n} &= \frac{\partial v_d}{\partial n} - i\frac{\partial v_{s-}}{\partial n} \\ &= \frac{\partial v_d}{\partial n} - i\left( \int_\gamma \phi(\mathbf{r_0}) \frac{\partial G(\mathbf{r}, \mathbf{r_0})}{\partial n(\mathbf{r})} d\gamma(\mathbf{r_0}) - \frac{1}{2}\phi(\mathbf{r}) \right) \\ &= -ik \end{aligned} \tag{4.15}$$

where this time we used the jump relation (4.10).

Now, $\frac{\partial v_d}{\partial n}$ is continuous, so we don't need to use a jump relation, however we haven't yet found an expression for it.

We define $\frac{\partial v_d}{\partial n}$ as follows:

$$\frac{\partial v_d}{\partial n} = k^2 \int_\gamma \phi(\mathbf{r_0}) G(\mathbf{r}, \mathbf{r_0}) d\gamma(\mathbf{r_0}) + \int_\gamma \phi(\mathbf{r_0}) \frac{\partial^2 G(\mathbf{r}, \mathbf{r_0})}{\partial y(\mathbf{r}) \partial y(\mathbf{r_0})} d\gamma(\mathbf{r_0}). \tag{4.16}$$

The proof of this is not so simple, and is outside the scope of this project, but, again, G. Chen and J. Zhou's *Boundary element methods with applications to nonlinear problems* explores this in more detail on pages 204–226 and 313 – specifically Theorems 6.3, 6.10 and 6.12 as well as equations 6.48, 6.49 and 7.43.

Thus, our BIE looks like:

$$\frac{i}{2}\phi(\mathbf{r}) = -ik + \int_\gamma \phi(\mathbf{r_0}) \left( -\frac{\partial^2 G(\mathbf{r}, \mathbf{r_0})}{\partial y(\mathbf{r}) \partial y(\mathbf{r_0})} + i\frac{\partial G(\mathbf{r}, \mathbf{r_0})}{\partial y(\mathbf{r})} - k^2 G(\mathbf{r}, \mathbf{r_0}) \right) d\gamma(\mathbf{r_0}), \tag{4.17}$$

where, again, we must solve for $\phi(\mathbf{r})$.

## 4.4 Boundary element methods

Now that we have our BIE, we look to solve it using the *boundary element method* (BEM), which obtains a numerical solution for $\phi$. Plugging this back into (4.9) gives us our numerical solution for $u^{(s)}$.

To begin, we discretise the boundary $\gamma$ into $N$ smaller plates, which we denote by $\gamma_i$, for $i = 1, \dots, N$. In our case, each *boundary element* $\gamma_i$ is the line $\{(x, 0) \in \mathbb{R}^2 : x \in [-1 + \frac{2(i-1)}{N}, -1 + \frac{2i}{N}]\}$.

In the centre of each boundary element we assign a *boundary node*, denoted by $x_i = (-1 + \frac{2i-1}{N}, 0)$. It is important that we do not place any nodes on either edge of our boundary, as the normal is not well-defined there, and it turns out that choosing our nodes in this manner ensures that the finite energy condition (2.6) is satisfied. In our case, we choose our nodes to be in the centre of each boundary element.

Our main assumption with this method is that on each boundary element $\gamma_i$, our weight function $\phi(\mathbf{r})$ is constant. That is, for each $\mathbf{r} \in \gamma_i$, $\phi(\mathbf{r}) = \tilde{\phi}(x_i) = \tilde{\phi}_i$. Obviously this is not necessarily true – however we pick $N$ to be sufficiently large compared to the wave number $k$ so that the correct behaviour is captured. In general, for boundary element methods, it is commonly accepted that setting $N = 10k$ is sufficient to have accurate behaviour both at the edges and in the middle of the obstacle.

This discretisation allows us to pull the $\phi(\mathbf{r_0})$ term out of the integral, which lets us turn the integral equation into a system of $N$ linear equations, which can easily be solved using matrices.

For the Dirichlet problem, we have:

$$\begin{aligned} \frac{1}{2}\tilde{\phi}(x_i) &= -1 + \sum_{j=1}^{N} \int_{\gamma_j} \tilde{\phi}(x) \left( iG(x_i, x) - \frac{\partial G(x_i, x)}{\partial y(x)} \right) d\gamma_j(x) \\ &= -1 + \sum_{j=1}^{N} \tilde{\phi}(x_j) \int_{\gamma_j} \left( iG(x_i, x) - \frac{\partial G(x_i, x)}{\partial y(x)} \right) d\gamma_j(x), \end{aligned} \tag{4.18}$$

where $i = 1, \dots, N$, where we used our assumption that $\tilde{\phi}(x) = \tilde{\phi}(x_j)$ for all $x \in \gamma_j$.

This then yields the equation

$$\frac{1}{2}\tilde{\phi}_i = -1 + \sum_{j=1}^{N} A_{ij}\tilde{\phi}_j, \tag{4.19}$$

where $A_{ij} = \int_{\gamma_j} \left( iG(x_i, x) - \frac{\partial G(x_i, x)}{\partial y(x)} \right) d\gamma_j(x)$.

Finally, this gives the matrix equation

$$B\tilde{\phi} = -1, \tag{4.20}$$

where $B = \frac{1}{2} I_N - A$, with $I_N$ being the $N \times N$ identity matrix.

This lets us solve for $\tilde{\phi}$, our piecewise-constant approximation for $\phi$, as long as we are able to calculate $A$.

When $i \neq j$, $A_{ij}$ can easily be computed numerically using a quadrature rule, however the case $i = j$ is tricky since we encounter a singularity at $x = x_i$. For this, we solve the integral analytically, as this integral is the same for all $i$.

Again, for the Dirichlet case,

$$\begin{aligned} A_{ii} &= \int_{\gamma_i} \left( iG(x_i, x) - \frac{\partial G(x_i, x)}{\partial y(x)} \right) d\gamma_i(x) \\ &= \int_{-\frac{1}{N}}^{\frac{1}{N}} \left( iG(0, x) - \frac{\partial G(0, x)}{\partial y(x)} \right) dx, \end{aligned} \tag{4.21}$$

where we use that $x_i$ lies in the centre of $\gamma_i$.

Now, using the definition of $\frac{\partial G(\mathbf{r},\mathbf{r_0})}{\partial n(\mathbf{r_0})}$ derived in (4.7), we can see that $\frac{\partial G(x_i,x)}{\partial y(x)} = 0 \quad \forall x \in \gamma_i$, as the $y$-components of both $x_i$ and $x$ are 0, and so $\mathbf{j} \cdot (x_i - x) = 0$.

So, our problem lies in calculating

$$\int_{-\frac{1}{N}}^{\frac{1}{N}} iG(0,x)dx = -\frac{1}{2}\int_{0}^{\frac{1}{N}} H_0^{(1)}(kx)dx. \tag{4.22}$$

We do this by using the small-$z$ approximation of $H_0^{(1)}(z)$:

$$H_0^{(1)}(z) \sim 1 + \frac{2i}{\pi}\left(\ln\left(\frac{z}{2}\right) + \gamma\right), \tag{4.23}$$

where $\gamma$ indicates the Euler-Mascheroni constant $\gamma \approx 0.57721...$, and $0 < z \ll 1$. [22]

Inputting this into (4.22) and solving the improper integral yields the solution

$$A_{ii} = -\frac{1}{2\pi N}\left(\pi + 2i\left(\ln\left(\frac{k}{2N}\right) + \gamma - 1\right)\right) \tag{4.24}$$

Now that we have calculated $A$, we may solve the matrix equation (4.20) for $\tilde{\phi}(\mathbf{r})$, which then in turn lets us find $u^{(s)}$ by plugging $\tilde{\phi}(\mathbf{r})$ into (4.9).

The Neumann case is a little more complicated – first we start with our BIE from (4.17), and discretise our boundary:

$$\begin{aligned}\frac{i}{2}\tilde{\phi}(x_i) &= -ik + \sum_{j=1}^{N}\int_{\gamma_j}\tilde{\phi}(x)\left(-\frac{\partial^2 G(x_i,x)}{\partial y(x_i)\partial y(x)} + i\frac{\partial G(x_i,x)}{\partial y(x_i)} - k^2 G(x_i,x)\right)d\gamma_j(x)\\ &= -ik + \sum_{j=1}^{N} A_{ij}\tilde{\phi}(x_j),\end{aligned} \tag{4.25}$$

where $A_{ij} = \int_{\gamma_j}\left(-\frac{\partial^2 G(x_i,x)}{\partial y(x_i)\partial y(x)} + i\frac{\partial G(x_i,x)}{\partial y(x_i)} - k^2 G(x_i,x)\right)d\gamma_j(x)$.

This then gives the matrix equation

$$B\tilde{\phi} = -ik, \tag{4.26}$$

where $B = \frac{i}{2}I_N - A$.

Now, when calculating our matrix $A$, again, we may solve the $i \neq j$ case numerically using a quadrature rule or otherwise, but the case $i = j$ still presents a problem due to the singularity at $x = x_i$, so we must solve this case analytically.

This time,

$$\begin{aligned}A_{ii} &= \int_{\gamma_i}\left(-\frac{\partial^2 G(x_i,x)}{\partial y(x_i)\partial y(x)} + i\frac{\partial G(x_i,x)}{\partial y(x_i)} - k^2 G(x_i,x)\right)d\gamma_i(x)\\ &= \int_{-\frac{1}{N}}^{\frac{1}{N}}\left(-\frac{\partial^2 G(0,x)}{\partial y(0)\partial y(x)} + i\frac{\partial G(0,x)}{\partial y(0)} - k^2 G(0,x)\right)dx.\end{aligned} \tag{4.27}$$

Again, by (4.7), $\frac{\partial G(x_i,x)}{\partial y(x_i)} = 0$ on $\gamma$, but this time we have the $\frac{\partial^2 G(x_i,x)}{\partial y(x_i)\partial y(x)}$ term that we have not yet found an expression for – however we can clearly see that this, too, will be 0 along $\gamma$,

since we already know that $\frac{\partial G(x_i,x)}{\partial y(x_i)} = 0$ on $\gamma$. This means that our analytical solution is the following:

$$A_{ii} = -\frac{ik^2}{2\pi N}\left(\pi + 2i\left(\ln\left(\frac{k}{2N}\right) + \gamma - 1\right)\right) \tag{4.28}$$

where we used our result from (4.24).

Now we have found a numerical approximation for $u^{(s)}(\mathbf{r})$, we can analyse how it looks close to the boundary as well as its behaviour as $||\mathbf{r}|| = r \to \infty$.

## 4.5 Verifying our solution

### 4.5.1 Near-field behaviour

In order to verify that our results make sense, we look more closely at what happens close to the obstacle – specifically what we expect to happen in a physical example of the problem.

By plotting our solution for $u^{(s)}$, we may verify that our solutions for both the Dirichlet and Neumann problems exhibit behaviour that we would expect by thinking about the problem in a physical sense. We can easily verify this by looking at Figures 7 and 8, and referring to the 4 sections outlined in 2 – indeed we do get a reflection region, two transmission regions, as well as a shadow region.

We also would like to confirm that the edge conditions (2.6) are satisfied – to do this we need only check that $u^{(s)}(\mathbf{r})$ grows sufficiently slowly as $\mathbf{r} \to (\pm 1, 0)$. Figures 9 and 10 verify this – thus we have behaviour we would expect in the near-field.

### 4.5.2 Far-field behaviour

As well as verifying the behaviour close to the obstacle, we also want to investigate what happens as we move away from the boundary. Our definition for the Sommerfeld radiation condition (2.5) states that as $||\mathbf{r}|| = r \to \infty$, $u^{(s)} \sim A(\theta)\frac{e^{ikr}}{r^{\frac{1}{2}}}$.

We can use this condition in order to sample $A(\theta)$ at points with varying $\theta$ for a fixed large $r$, by sampling $u^{(s)}(\mathbf{r})$ for points of the form $\mathbf{r} = (R\cos\theta, R\sin\theta)$, with $R \gg 1$ chosen to be large, and $\theta$ varying from $0$ to $2\pi$. By doing this, we use that $u^{(s)} \sim A(\theta)\frac{e^{ikR}}{R^{\frac{1}{2}}}$ to sample $A(\theta)$.

Plotting this will then give us an indication of the far-field behaviour of our result – and we may investigate it to see whether it yields a diffraction pattern as we may expect.

Figures 11 and 12 show the results obtained from this procedure for the Dirichlet and Neumann conditions respectively – and we can clearly see that the results aren't too different from each other. This is to be expected, as in the far-field we are just seeing the diffraction pattern generated from the obstacle, which is almost the same regardless of whether the object is sound-soft or sound-hard.

### 4.5.3 Mesh Convergence Analysis

We also use a mesh convergence study to verify the accuracy of our numerical solution. As demonstrated by Atkinson and Chandler [23], the accuracy of a numerical solution is fundamentally linked to the *mesh size*, denoted as $h$. While our BEM differs from the collocation

method analysed in their work, the underlying principle that a finer discretisation yields a more accurate solution remains the same.

We calculate the far-field amplitude, $A(\theta)$, for a fixed angle $\theta$ and wave number $k$ over a range of discretisation levels, $N$. By observing the behavior of $|A(\theta)|$ as $N$ grows, we can numerically verify the solver's convergence.

Figures 13 and 14 show the results of this analysis. In Figure 13, the plot of the calculated amplitude versus the number of elements shows the value stabilizing and "petering off" as $N$ becomes large indicating convergence. This behavior is analogous to the error tables presented by Atkinson and Chandler, where the error consistently decreases and the solution's accuracy improves as the number of elements is increased.

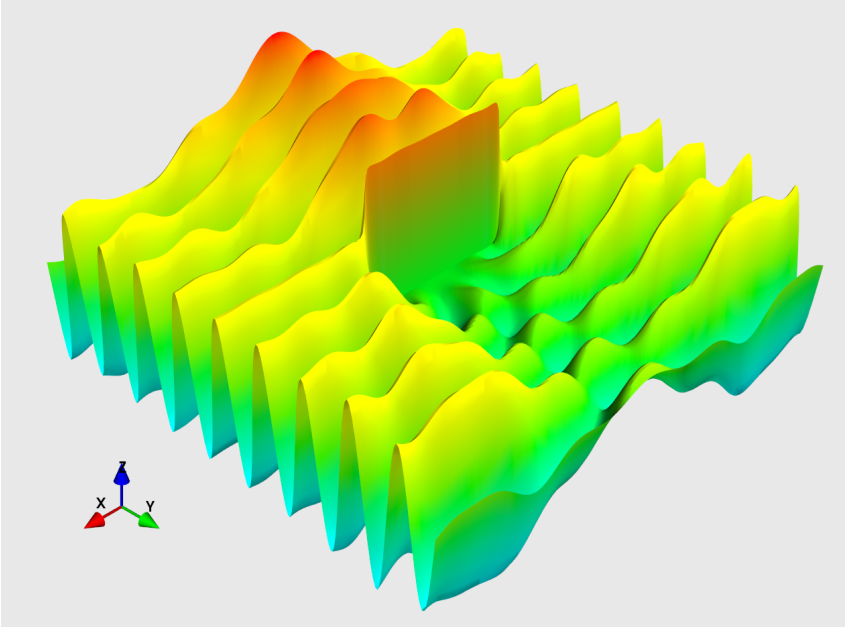



**Figure 7** Real part of the total solution $u(x, y)$ to the Dirichlet problem with $k = 10$, sampled on a $200 \times 200$ grid over $\{(x, y) : -3 \leq x \leq 3, -3 \leq y \leq 3\}$. Notice the reflection of the wave occuring at $y < 0$ as well as the shadow region and diffraction occuring at $y > 0$.

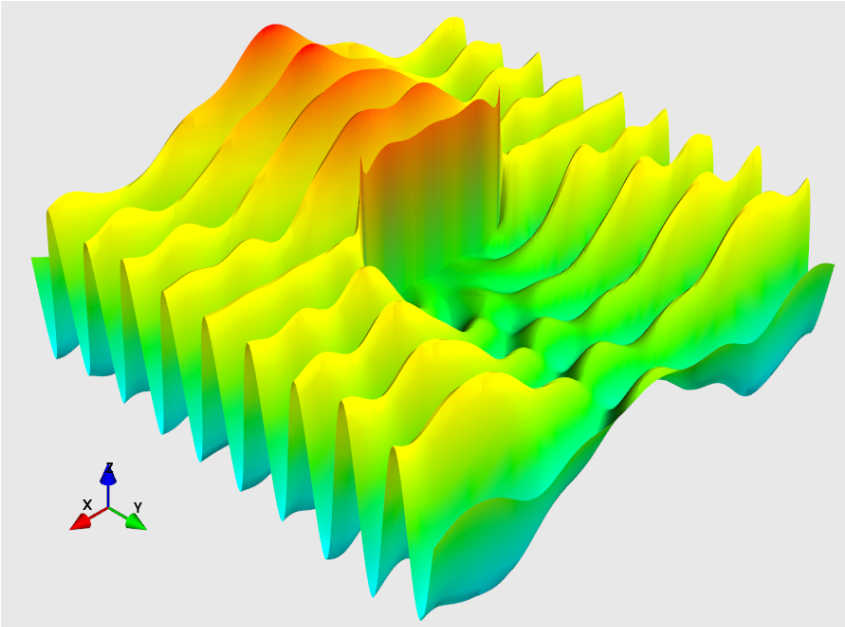



**Figure 8** Real part of the total solution $u(x, y)$ to the Neumann problem with $k = 10$, sampled on a $200 \times 200$ grid over $\{(x, y) : -3 \leq x \leq 3, -3 \leq y \leq 3\}$. Notice the similarity to the Dirichlet case – this is to be expected as the only difference is whether the obstacle is considered *'sound-soft'* or *'sound-hard'*.

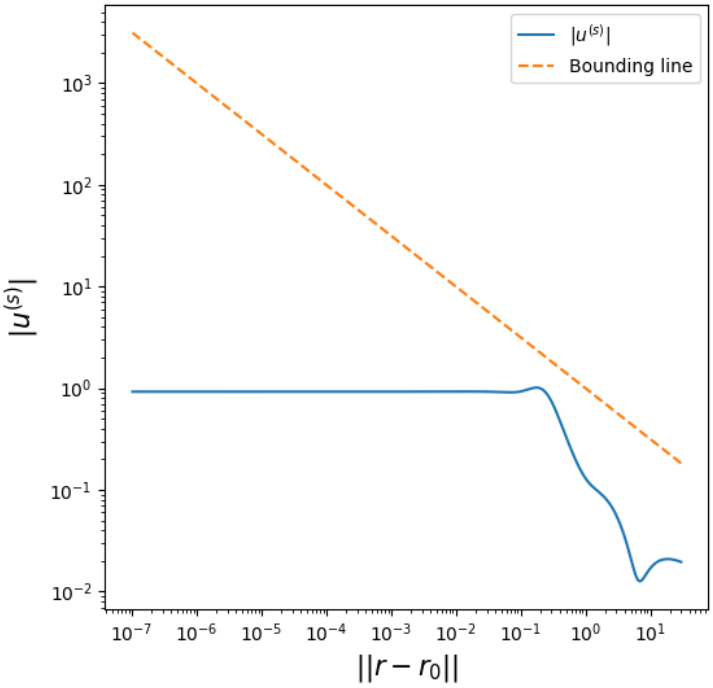



**Figure 9** Log-log plot of the amplitude of the Dirichlet solution $|u^{(s)}(\mathbf{r})|$ against $||\mathbf{r} - \mathbf{r_0}||$, with the upper bound of $||\mathbf{r} - \mathbf{r_0}||^{-\frac{1}{2}}$ plotted for comparison. For this graph, $\mathbf{r_0} = (1, 0)$, $\mathbf{r}$ lie on the half-line $\{(x, y) : y = 1 - x,\ x > 1\}$, $k = 25$, and we sample $1000$ points.

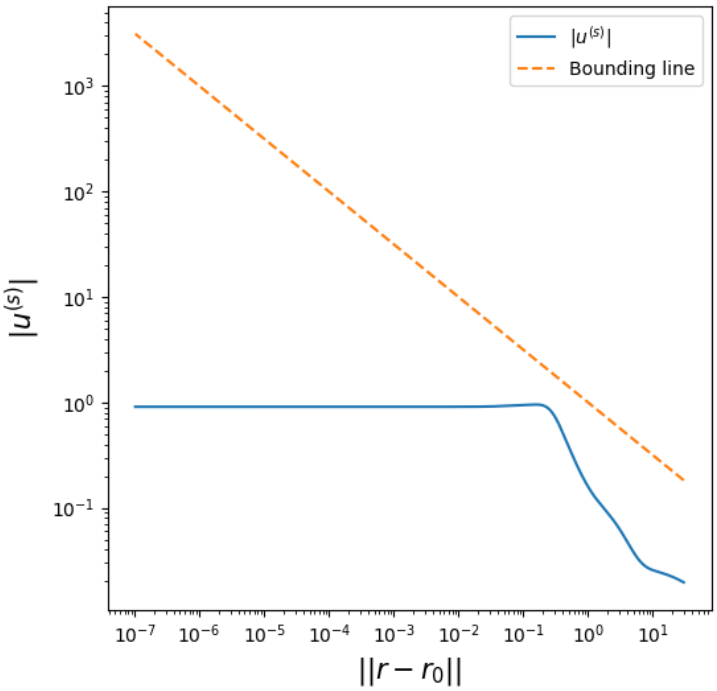



**Figure 10** Log-log plot of the amplitude of the Neumann solution $|u^{(s)}(\mathbf{r})|$ against $||\mathbf{r} - \mathbf{r_0}||$, with the upper bound of $||\mathbf{r} - \mathbf{r_0}||^{-\frac{1}{2}}$ plotted for comparison. For this graph, $\mathbf{r_0} = (1, 0)$, $\mathbf{r}$ lie on the half-line $\{(x, y) : y = 1 - x,\ x > 1\}$, $k = 25$, and we sample $1000$ points.

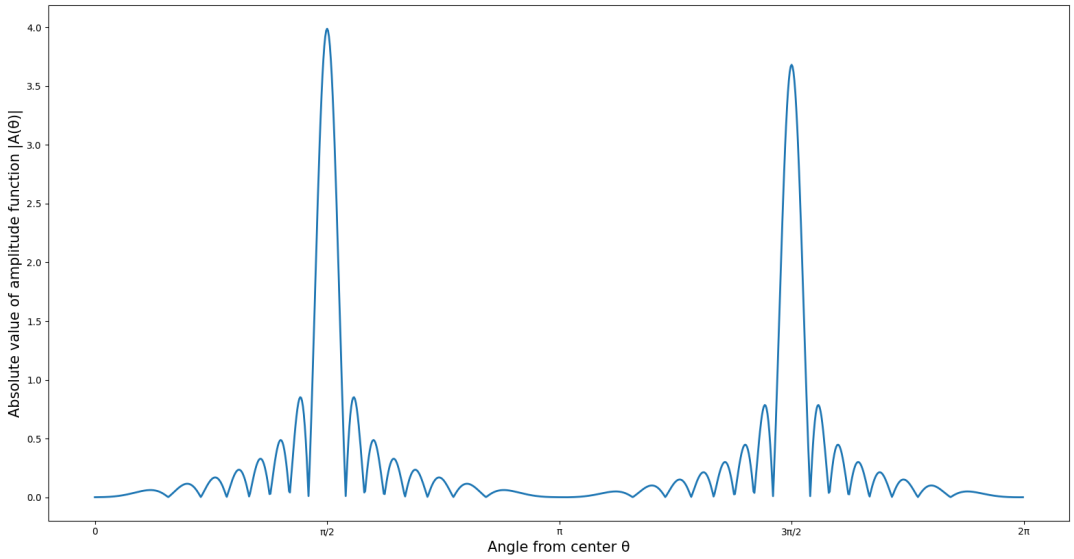



**Figure 11** Plot of $|A(\theta)|$ against $\theta$, the angle from the centre, in $u^{(s)} = A(\theta)\frac{e^{ikr}}{r^{\frac{1}{2}}}$, using $1000$ samples with $r = 10^7$ and $k = 25$, under Dirichlet conditions.

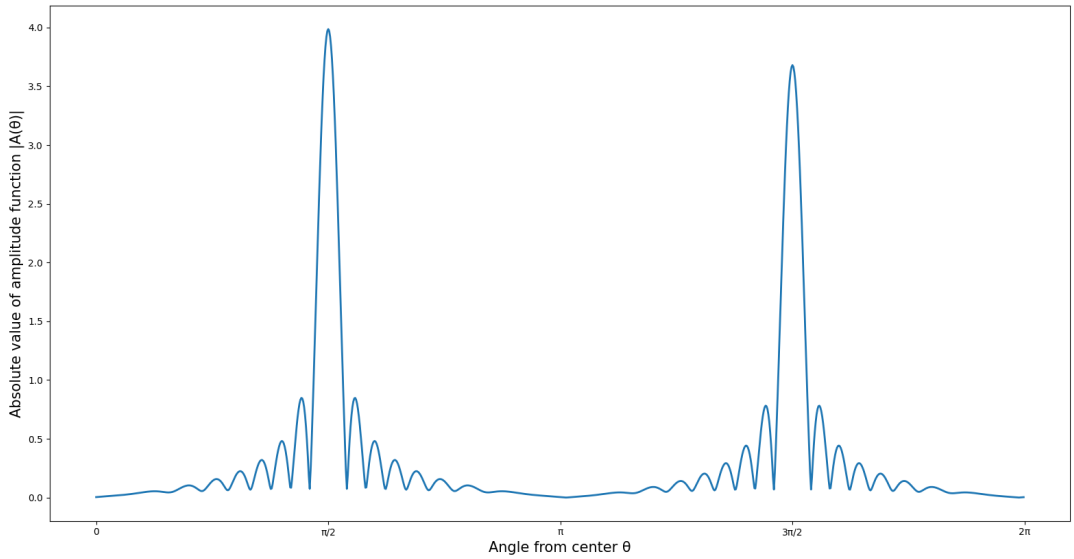



**Figure 12** Plot of $|A(\theta)|$ against $\theta$, the angle from the centre, in $u^{(s)} = A(\theta)\frac{e^{ikr}}{r^{\frac{1}{2}}}$, using $1000$ samples with $r = 10^7$ and $k = 25$, under Neumann conditions.

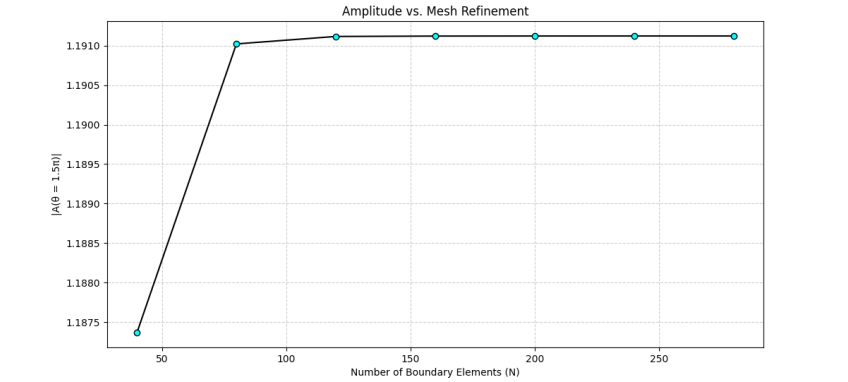



**Figure 13** A plot of the far-field amplitude magnitude $|A(\theta)|$ against the number of boundary elements $N$, for fixed $k = 5$ and $\theta = \pi/4$. The amplitude clearly converges to a stable value as $N$ increases.

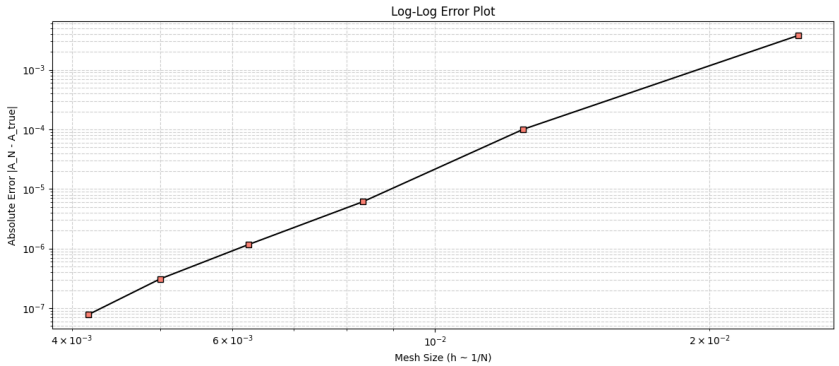



**Figure 14** A log-log plot of the absolute error in the far-field amplitude against the element size $(1/N)$. The error is calculated as the difference from a high-fidelity solution with $N = 2000$. The straight-line relationship indicates a clear polynomial order of convergence.

## 4.6 Refinements and extensions to the formulation

There are lots of ways to refine and generalise the model, which will be detailed in the following subsection. For simplicity, we will just stick to the *sound-hard* case, which lends itself better to acoustic problems – however similar procedures can be taken for the *sound-soft* case as in Section 4.3.1.

### 4.6.1 Solving for an arbitrary plate

To consider much more general settings (for example with multiple plates) we will parameterise the boundary as an interval $[0, 1]$.

Suppose that a boundary is defined between the points $\mathbf{a} = (a_1, a_2)$ and $\mathbf{b} = (b_1, b_2)$. We parameterise this as

$$\mathbf{r_0}(t) = \mathbf{a} + t(\mathbf{b} - \mathbf{a}), \quad t \in [0, 1] \tag{4.29}$$

After we define the following:

$$ds = ||\mathbf{r_0}'(t)||dt = ||\mathbf{b} - \mathbf{a}||dt = Ldt \tag{4.30}$$

$$\mathbf{n} = \frac{1}{L}\begin{pmatrix} -(b_2 - a_2) \\ b_1 - a_1 \end{pmatrix} \tag{4.31}$$

We can use the above definitions and (4.10) to get

$$\frac{i}{2}\phi(\mathbf{r}) = -\frac{\partial u^{(i)}}{\partial n(\mathbf{r})} + \int_\gamma \phi(\mathbf{r_0}) \left( -\frac{\partial^2 G(\mathbf{r}, \mathbf{r_0})}{\partial y(\mathbf{r}) \partial y(\mathbf{r_0})} + i\frac{\partial G(\mathbf{r}, \mathbf{r_0})}{\partial y(\mathbf{r})} - k^2 G(\mathbf{r}, \mathbf{r_0}) \right) d\gamma(\mathbf{r_0}), \tag{4.32}$$

After these redefinitions, we may proceed as usual as in 4.4.

Once we have a single plate which can rotate and be placed anywhere in 2D space, we can very easily add more plates to create complex geometries to demonstrate interesting wave properties such as single-slit diffraction. Suppose we index each plate as $\gamma_p$ and then we can define the new boundary $\gamma$ as:

$$\gamma = \bigcup_{p=1}^{P} \gamma_p$$

Figures 15 and 16 show the results of different applications of this adjustment, such as modelling diffraction from an angled plate, or using multiple plates to model single-slit diffraction.

### 4.6.2 Solving for an arbitrary plane wave

So far we have only considered a simple version of the wave equation ($u^{(i)}(x, y) = e^{iky}$) to be normally incident on a horizontal plate. Now we consider the general wave equation

$$u^{(i)}(\mathbf{r}) = e^{ik\mathbf{d}\cdot\mathbf{r}} = e^{ik(x\cos\alpha + y\sin\alpha)} \tag{4.33}$$

where $\alpha$ is the angle of incidence with respect to the positive x-axis. It is a simple exercise to show it still satisfies the Helmholtz equation, as $\nabla^2 u^{(i)} = -k^2|\mathbf{d}|^2 u^{(i)} = -k^2 u^{(i)}$. In the Neumann problem, we are concerned with: $\frac{\partial u^{(s)}}{\partial n} = -\frac{\partial u^{(i)}}{\partial n}$ so we need to recalculate $\frac{\partial u^{(i)}}{\partial n}$. The gradient of $u^{(i)}$ is

$$\nabla u^{(i)}(\mathbf{r}) = ik(\cos\alpha, \sin\alpha)e^{ik(x\cos\alpha + y\sin\alpha)} = ik\mathbf{d}u^{(i)}(\mathbf{r}) \tag{4.34}$$

Thus for any point $\mathbf{r} \in \gamma$

$$\frac{\partial u^{(i)}}{\partial n_r}(\mathbf{r}) = \nabla u^{(i)}(\mathbf{r}) \cdot \mathbf{n}_r = ik(\mathbf{d} \cdot \mathbf{n}_r)u^{(i)}(\mathbf{r}) \tag{4.35}$$

where $\mathbf{n}_r$ is the normal vector at $\mathbf{r}$. If we now consider the discretised system, where $\mathbf{r}_i = (r_{i,1}, r_{i,2})$ is the coordinates of the midpoint $i$-th element on the boundary, then

$$g_i' = -\frac{\partial u^{(i)}}{\partial n_{r_i}}(\mathbf{r}_i) = -ik(\mathbf{d} \cdot \mathbf{n}_{r_i})e^{ik(r_{i,1}\cos\alpha + r_{i,2}\sin\alpha)} \tag{4.36}$$

Figure 17 shows an example of this in use – as we can see, we get the desired behaviour behind the obstacle.

### 4.6.3 Refining the discretisation

The discretisation can be done simply by dividing the boundary uniformly, as in Section 4.4, or it can be adjusted to be more irregular. One option is to divide the $\gamma$ into three regions: one region on each edge and one in the middle. Sharp features like edges are the source of diffraction patterns, as was seen in Section 3.3, so the majority of the boundary divisions should be around these edges. Empirically, $5k$ divisions in a radius of $1/k$ around each corner is commonly accepted to be sufficient for our purposes. We choose $1/k$ since this allows the model to capture the behaviour of the wave length on the order of $1/k$ well. We then place an arbitrarily small number of divisions in the middle portion of the boundary, as behaviour there is less crucial. Our approach in this regard is implemented in the `interval_creator` function in the file `indirect_BEM_combined.py` provided in the repository (see appendix A).

We now discuss a potential issue with discretising on the boundary. Recall from Section 4.2 the definition of Green's function:

$$G(\mathbf{r}, \mathbf{r_0}) = \frac{i}{4}H_0^{(1)}(k||\mathbf{r} - \mathbf{r_0}||) \tag{4.37}$$

where

$$H_0^{(1)}(z) \sim 1 + \frac{2i}{\pi}\left(\ln\left(\frac{z}{2}\right) + \gamma\right) \tag{4.38}$$

So as $||\mathbf{r}_i - \mathbf{r_0}|| \to 0$ then the logarithm term tends to $-\infty \implies$ the integral is improper. This was handled analytically in section 4.4 using the small-$z$ approximation for $H_0^{(1)}(z)$, however, an alternative method is to define a *second auxiliary boundary* slightly seperated from the original interval. This approach is called the *auxiliary method*.

Empirically, a good distance between $\gamma$ and $\gamma_{aux}$ is $\delta$, with:

$$\delta = 0.1\frac{L}{N} \tag{4.39}$$

where $L$ is the length of $\gamma$ and $N$ is the number of discretisations on $\gamma$.

Now we must map every point $\mathbf{r_0}(t)$ on $\gamma$ to $\gamma_{aux}$ by

$$\mathbf{r}_{0,aux}(t) = \mathbf{r_0}(t) - \delta\mathbf{n}(t) \tag{4.40}$$

This $\mathbf{r}_{0,aux}$ is used in place of $\mathbf{r_0}$ in the algorithm. It turns out that by using this auxiliary boundary, we capture the jump relations, since instead of integrating over the boundary, we are integrating in the limit as we approach the boundary – which is precisely what the jump relations account for.

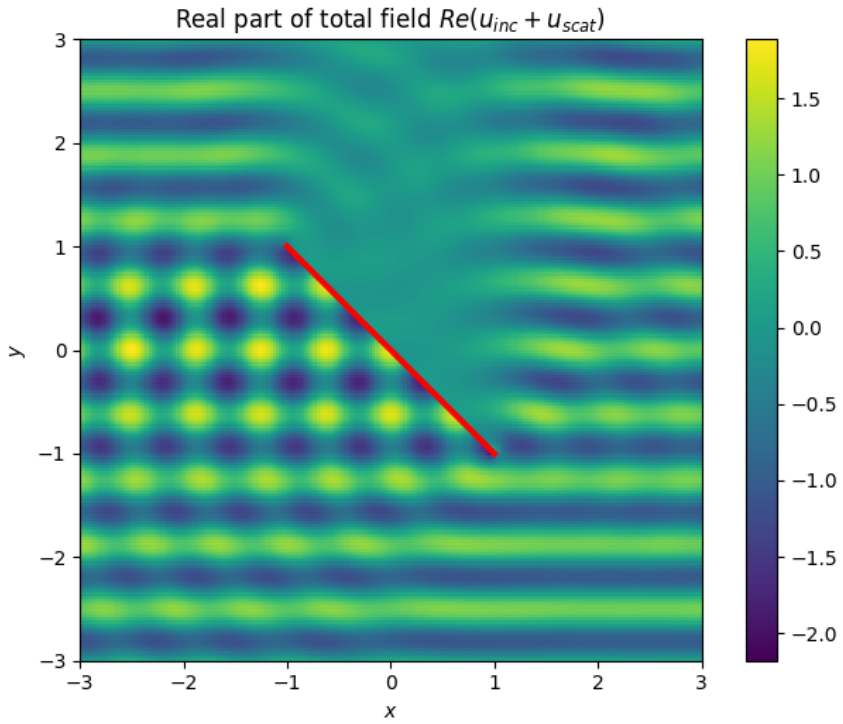



**Figure 15** Real part of the total solution $u(x, y)$ to the Neumann problem with $k = 10$, $u^{(i)} = e^{iky}$, $\gamma = \{(x, y) : y = -x, -1 \leq x \leq 1\}$, sampled on a $200 \times 200$ grid over $\{(x, y) : -3 \leq x \leq 3, -3 \leq y \leq 3\}$.

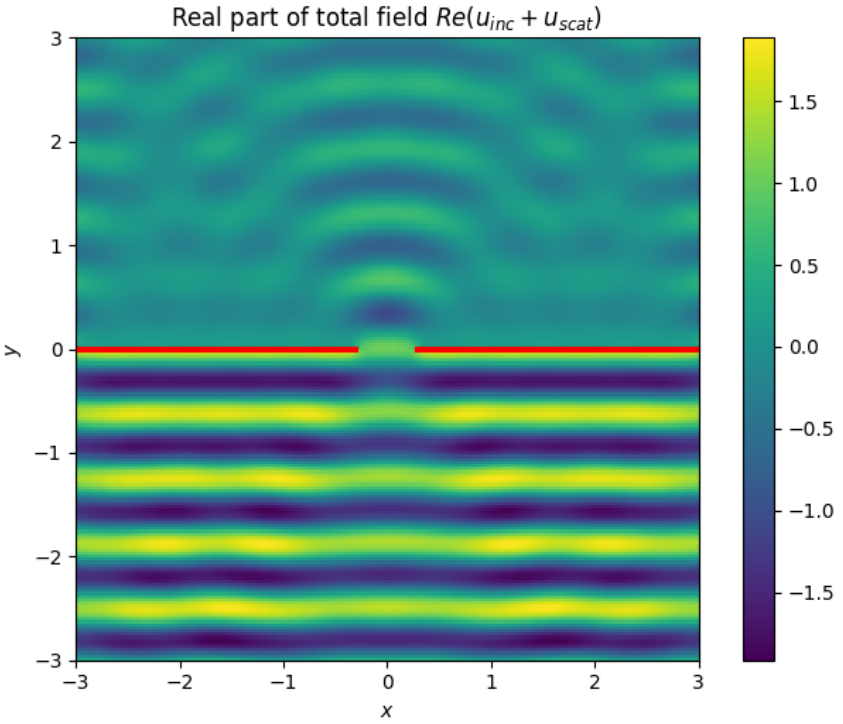



**Figure 16** Real part of the total solution $u(x, y)$ to the Neumann problem with $k = 10$, $u^{(i)} = e^{iky}$, $\gamma = \{(x, y) : y = 0, -3 \leq x \leq -0.3\} \cup \{(x, y) : y = 0, 0.3 \leq x \leq 3\}$, sampled on a $200 \times 200$ grid over $\{(x, y) : -3 \leq x \leq 3, -3 \leq y \leq 3\}$.

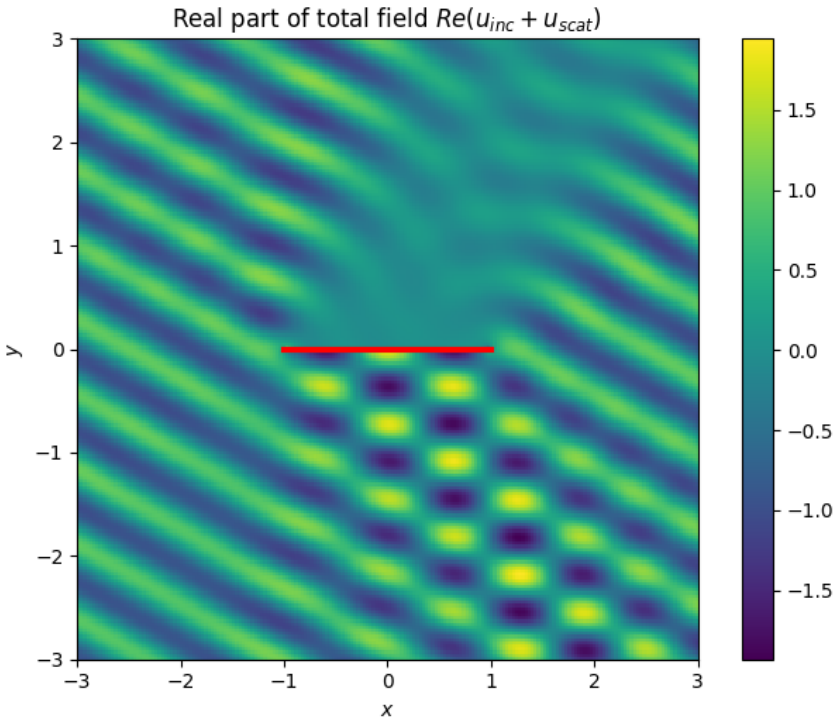



**Figure 17** Real part of the total solution $u(x, y)$ to the Neumann problem with $k = 10$, $u^{(i)} = e^{ik(x\cos\frac{\pi}{3}+y\sin\frac{\pi}{3})}$, $\gamma = \{(x, y) : y = 0, -1 \leq x \leq 1\}$, sampled on a $200 \times 200$ grid over $\{(x, y) : -3 \leq x \leq 3, -3 \leq y \leq 3\}$.

# 5 Comparison and conclusion

## 5.1 Comparison

The results gathered from our two methods provide different insights into the problem. The asymptotic solution, through the use of the GTD together with the matched Sommerfeld solution, derived a leading-order approximation for the large-$k$ asymptotic form of our finite-plate problem. By analysing the diffracted rays, we can compare our results to our numerical solution.

While our numerical solution is limited in regards to the large-$k$ limit – the computation time increases at an order of around $O(k^3)$ as we increase $k$ – we can compare our results in the far-field. Using the Sommerfeld radiation condition (2.5), we can again inspect the amplitude function $A(\theta)$ as $\theta$ changes. Figures 11 and 12 showed clear peaks at $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$, as we may expect, and this result is further supported by the results found from the analytical method – taking a close look at the form of the solution wave shows peaks in those same places, representing the part of $u^{(s)}$ that is reflected from the boundary and the part that cancels out $u^{(i)}$ behind the boundary, for $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$ respectively.

## 5.2 Conclusion

This project has investigated and solved the problem of acoustic plane wave diffraction on a finite, two-dimensional plate. We approached the problem in two different ways: asymptotically with geometric theory of diffraction (GTD) and then numerically with the boundary element method.

The asymptotic approach solved the problem by first considering the Sommerfeld Problem of diffraction on a semi-infinite plate. The analytical solution was found using parabolic-cylindrical coordinates and expressed in terms of Fresnel integrals. To adapt the problem back to the finite plate, we utilised the GTD and matched asymptotic expansion for high-frequency waves.

Finally, we solved both the amplitude and phase function to obtain the aymptotic solution for both Dirichlet and Neumann conditions.

The numerical approach utilised the indirect boundary element method. We first represented the scattered wave as a continuous distribution of potentials across the plate. In this project we used both single and double layer potentials which came from Green's function and it's partial derivatives. We then imposed boundary conditions (Dirichlet or Neumann) and using jump relations for layer potentials, formulated a boundary integral equation. We solved this integral by discretising the boundary and using numerical quadrature on each boundary element.

The numerical solution successfully produced results for both Dirichlet (sound-soft) and Neumann (sound-hard) conditions. Additionally, we verified the near-field and far-field conditions by comparing it to plots that we would expect from existing literature. These showed the expected diffraction patterns. We also used a mesh convergence analysis which provides strong numerical evidence for the accuracy of the BEM implementation.

We considered an extension to our numerical solution to show how it could be generalised to more complex geometries, such as those containing slits or angled plates. Also, we discussed ways that could make our method more accurate such as using the auxiliary method or using non-uniform plate discretisation. Our results demonstrate the robustness of BEM for solving a wide range of acoustic problems.

This work leaves some areas still to be explored; the problem could be generalised to three

dimensions, allowing it to be applied to real-world obstacles. Furthermore, we currently only consider geometries generated by straight line plates but this could be extended to curved surfaces. There is also room for optimisation in our numerical implementation, as the algorithm's complexity is of high polynomial order, increasing the computation time greatly when the number of elements per boundary is specified to be very high or when the waves are high-frequency, which forces the number of elements per boundary to be high. This would allow for better comparison between the analytical and numerical approaches, as we could gather results for larger values of $k$.

# A Appendix

All plots and figures displayed in this paper (besides Figure 4, see [18]) as well as code used to solve the problem numerically were created by us – all our resources are available publicly at https://github.com/typtrue/m2r-project.

## References


[1] F. Grimaldi. *Physico-mathesis de lumine, coloribus, et iride, aliisque adnexis libri duo: opus posthumum*, page 2. 1665.

[2] D. A. der Wissenschaften zu Berlin. *Histoire de l'Académie royale des sciences et des belles-lettres de Berlin: avec les mémoires pour la même année, tirez des registres de cette Academie*, pages 214–219. Ambroise Haude, 3rd edition, 1749.

[3] D. J. Griffiths and D. F. Schroeter. *Introduction to Quantum Mechanics*. Cambridge University Press, 3rd edition, 2018.

[4] A. Sommerfeld. Mathematische theorie der diffraction. *Mathematische Annalen*, 47(2): 317–374, Jun 1896.

[5] H. Lamb. On sommerfeld's diffraction problem; and on reflection by a parabolic mirror. *Proceedings of the London Mathematical Society*, s2-4(1):190–203, 01 1907.

[6] J. B. Keller. Geometrical theory of diffraction$*$. *J. Opt. Soc. Am.*, 52(2):116–130, Feb 1962.

[7] G. Green. *An Essay on the Application of Mathematical Analysis to the Theories of Electricity and Magnetism*. author, 1828.

[8] G. Chen and J. Zhou. *Boundary element methods with applications to nonlinear problems*. Atlantis studies in mathematics for engineering and science, v. 7. Atlantis Press, Amsterdam, 2nd edition, 2010.

[9] D. Colton and R. Kress. *Integral Equation Methods in Scattering Theory*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 2013.

[10] B. Noble. *Methods Based on the Wiener-Hopf Technique for the Solution of Partial Differential Equations*. AMS Chelsea Publishing Series. Chelsea Publishing Company, 1958.

[11] D. Crighton, A. Dowling, J. Williams, M. Heckl, and F. Leppington. *Modern Methods in Analytical Acoustics: Lecture Notes*. Springer London, 1992.

[12] M. Born, E. Wolf, and A. Bhatia. *Principles of Optics*. Cambridge University Press, 7th edition, 2019.

[13] J. Meixner. Die kantenbedingung in der theorie der beugung elektromagnetischer wellen an vollkommen leitenden ebenen schirmen. *Annalen der Physik*, 441(1):1–9, 1950.

[14] K. T. McDonald. Sommerfeld's diffraction problem. http://kirkmcd.princeton.edu/examples/sommerfeld.pdf, 2019. Accessed 14th June 2025.

[15] C. Balanis. *Advanced Engineering Electromagnetics*. CourseSmart Series. Wiley, 2012.

[16] H. Lamb. On sommerfeld's diffraction problem; and on reflection by a parabolic mirror. *Proceedings of the London Mathematical Society*, s2-4(1):190–203, 01 1907.

[17] Wikipedia contributors. Gaussian integral — Wikipedia, the free encyclopedia, 2025. [Online; accessed 14th June 2025].

[18] J. B. Keller. Rays, Waves and Asymptotics. *Bulletin of the American Mathematical Society*, 84(5):727–750, 1978.

[19] DLMF. *NIST Digital Library of Mathematical Functions*. https://dlmf.nist.gov/10.17, Release 1.2.4 of 2025-03-15. F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.

[20] S. Chandler-Wilde and S. Langdon. Boundary element methods for acoustics. *Lecture notes, University of Reading, Department of Mathematics*, 2007.

[21] H. Brakhage and P. Werner. Über das dirichletsche außenraumproblem für die helmholtzsche schwingungsgleichung. *Archiv der Mathematik*, 16(1):325–329, Dec 1965.

[22] M. Abramowitz and I. A. Stegun. *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. Dover, New York, ninth dover printing, tenth gpo printing edition, 1964.

[23] K. Atkinson and G. Chandler. The collocation method for solving the radiosity equation for unoccluded surfaces. *Journal of Integral Equations and Applications*, 10(3):253–290, 1998.